Fibrational Semantics for Coexisting Implications

Maria Emilia Maietti[*] Valeria de Paiva[‡] Eike Ritter[†]

[*]Dipartimento di Matematica, Università di Padova, Italy. maietti@math.unipd.it
[†]School of Computer Science, University of Birmingham, UK. E.Ritter@cs.bham.ac.uk
[‡]Topos Institute, Berkeley CA, USA. valeria@topos.institute

Abstract

This paper provides a comprehensive fibrational categorical semantics for three type theories that combine intuitionistic and linear logic: ILT (Intuitionistic and Linear Type Theory), DILL (Dual Intuitionistic Linear Logic of Barber and Plotkin), and LNL (Linear-Non-Linear calculus of Benton).

ILT has both intuitionistic implication ($A \to B$) and linear implication ($A \multimap B$) as primitive operations, with no modality !. This makes its categorical semantics challenging: the standard monoidal adjunction models of DILL and LNL are defined precisely to accommodate the modality, and cannot be directly restricted to ILT. The key idea is to *linearise* the fibrational semantics of the simply-typed λ-calculus: we replace the cartesian fibres of the standard D-category construction with symmetric monoidal closed fibres. The resulting *ILT-indexed categories* give a sound and complete semantics, and ILT is their internal language.

We extend this fibration framework to DILL and LNL, and prove these fibrational models equivalent to the classical monoidal-adjunction models. A further contribution is a conservativity theorem: DILL is a conservative extension of ILT, and the types outside ILT are precisely those of the form $A \multimap !B$ and $!!A$. This justifies the use of ILT in applications of linear logic to functional programming, where the modality is often unnecessary.

The fibrational perspective also illuminates the path toward *dependent* extensions of these systems: the comprehension adjunction and fibred SMC structure already present in ILT-indexed categories provide the semantic infrastructure needed to add dependent intuitionistic types, pointing toward a dependent ILT whose relationship to recent work on linear dependent type theory we discuss. A short version of this paper appeared as [MdPR00].

1 Introduction

Linear Logic and linear typing systems have been very influential in the more than 30 years since Girard first described the logical system [Gir87a]. Linear logic captured the imagination of computer scientists, who envisaged all kinds of applications. Many of these applications have been investigated, several implemented, but it is fair to say that many have not lived up to expectactions. A possible reason would be the lack of modularity of linear systems.

This paper describes work on a different linear system, introduced with an intended generalization to the Calculus of Constructions and corresponding implementation planned, but not developed at the time, due to moving personnel. This system would help not only with the modularity issue, but possibly also with other combinations of systems. We introduce a new term calculus for a logic that combines intuitionistic linear logic with intuitionistic logic. Since there are already several such logics (and corresponding term calculi) in the literature, we start by explaining what is our motivation for yet another such calculus.

2 Linear Lambda Calculi

The existence of several linear typing systems owes much to a mismatch, originally pointed out by Wadler, between the linear syntax and its proposed semantics. Under the humourous title of 'There is no substitute for Linear Logic' [Wad91], Wadler described how the syntax for the linear lambda calculus described by Abramsky did not agree with the categorical semantics for the same system, originally described by Seely. Thus there was clearly interest in obtaining a linear lambda calculus that could be proved Curry-Howard isomorphic to intuitionistic linear logic. Eventually the problem, which was lack of substitutivity for the linear lambda-calculus as far as the rules for "!" were concerned, was solved, using guidance from the categorical modelling of the system. We call the first system proved correct ILL ([BBdH93b, Bie94b, BBdH93a, Bie94a]).

The categorical modelling used for ILL seemed very convoluted, in particular there are many non-universal categorical constructions associated with the calculus. Benton [Ben95] embarked on a programme to simplify it, at the same time developing another linear lambda calculus, to match the simplified categorical semantics. Thus the calculus LNL (for Linear-Non-Linear lambda-calculus) and its semantics made its first appearance.

However, the LNL calculus looked very much like category theory passing itself as type theory. The traditional features of functional syntax were missing and hence more or less simultaneously with the appearance of LNL, Barber and Plotkin proposed another linear lambda-calculus which kept the simplified semantics of LNL, improved dramatically on the complexity of terms of ILL and for which good syntactic properties could be proved. The reason for the good syntactic properties is that the term constructors follow the standard pattern of introduction and elimination rules. Plotkin and Barber called their calculus DILL, for Dual Intuitionistic and Linear

Logic [Bar97].

Like LNL, the DILL calculus took the view that linear logic and intuitionistic logic should be seen as complementary, as opposed to the older Girard view that intuitionistic logic was to be decomposed in terms of a more primitive logic, linear logic. This change in perspective also matched Girard's discussions on the unity of logic and its formalization as LU [Gir93].

By then (around 1996) we had three linear term calculi (ILL, LNL, DILL), each with sound and complete categorical semantics, all proved Curry-Howard isomorphic to intuitionistic linear logic. More precisely, the set of types A for which there is a term M such that $\_ \vdash M : A$ (and which therefore correspond via Curry-Howard to provable theorems) is the same for all three term calculi. Moreover, there was some evidence that the different proof systems themselves were "essentially" equivalent: each and every proof in each of the systems could be translated into each of the other systems. Given this superficial evidence no one bothered much to straighten the situation. To some extent it seemed that choosing between these calculi was a matter of application at hand: if you wanted to know *where* contraction and erasing needed to occur in your term and you did not mind long/complex terms you should use ILL. If you preferred your semantics as neat and easy as possible and did not mind logical operators that looked like functors F and G, an adjunction in disguise, you could use LNL. And if you wanted the simplified semantics, but also wanted a term-calculus syntax that embedded the traditional simply typed lambda-calculus and preferred less complicated terms you should use DILL.

There would be nothing bothersome about this situation, if we could prove that "essentially" equivalent semantics really were equivalent, in some (categorical) sense of equivalence.

3 Implementing Linear Calculi

Assuming DILL was the most appropriate calculus to implement a version of linear functional programming, we set off to do it. However, while implementing DILL, we realized that more than the syntactic properties that had been proved for DILL, we needed some clear-cut separation between the linear functions and the intuitionistic ones.

In DILL intuitionistic variables exist independently from linear ones (this is the meaning of the dual context) but functions are all linear: to create an intuitionistic function one uses Girard's translation i.e. one remembers that $A \to B := (!A) \multimap B$. So you transform a linear variable into an intuitionistic one and then abstract over it.

From the implementation point-of-view it made much more sense to have both implications (linear $\multimap$ and intuitionistic $\to$) as primitive operations, side-by-side, and to give up on the modality !, source of several complications. The main reason is that the implementation of this system can be done in a modular way: One starts with any implementation of the λ-calculus (for the connectives $\to$ and $\times$), and adds to it an implementation of the linear part of the calculus, namely tensor products and linear implications. As a consequence, optimisations based on linearity can be restricted to a separate module, and do not interfere with the implementation of non-linear terms to which these optimisations do not apply. Such a separation of terms into linear and non-linear terms via a term's outermost connective is not available in DILL, hence a similar modularisation is more difficult to achieve.

Another (more proof-theoretical) reason is that in the translation of the simply-typed λ-calculus into DILL the modality ! occurs only in types $!A \multimap B$, and for these linearity is in effect not used. Indeed, a function of this type is applied only to arguments with no free linear variables, and during normalization of the term (execution of the program) these arguments will be substituted only for intuitionistic variables. Moreover, we want to detect immediately when a function is intuitionistic.

Thus instead of implementing DILL we ended up implementing a different linear typing system, anticipated by Plotkin [Plo93] and Wadler [Wad90].

4 The system ILT

The system ILT seems a good design choice, one that more people should know about, because it possesses the best kind of categorical semantics we can provide: an internal language categorical semantics.

As an intuitionistic and linear type theory, the syntax of ILT is easy. But we were confronted with the problem of providing categorical semantics for the ILT system. If the syntactic behaviour of ILT is very similar to that of DILL, when it comes to semantics, the situation is more complicated. It is not obvious how to restrict the idea of a *symmetric monoidal adjunction*, so that we capture all the behaviour of intuitionistic implication, without at the same time, importing all the machinery needed for modelling the modality !.

The solution of the categorical semantics puzzle came by looking at our previous models for calculi of explicit substitution in typed λ-calculi, which are models for intuitionistic logic [GdPR99]. These models involve fibrations in an essential way: the base category models contexts, and the fibres model terms-in-context. Substitution is then modelled by re-indexing (see Figure 1). Such a fibration takes the view that terms are parametrised by the variables occurring in them.

The categorical model for ILT presented in this paper combines modelling intuitionistic logic using fibrations with modelling (intuitionistic) linear logic using symmetric monoidal closed categories, and mostly important,

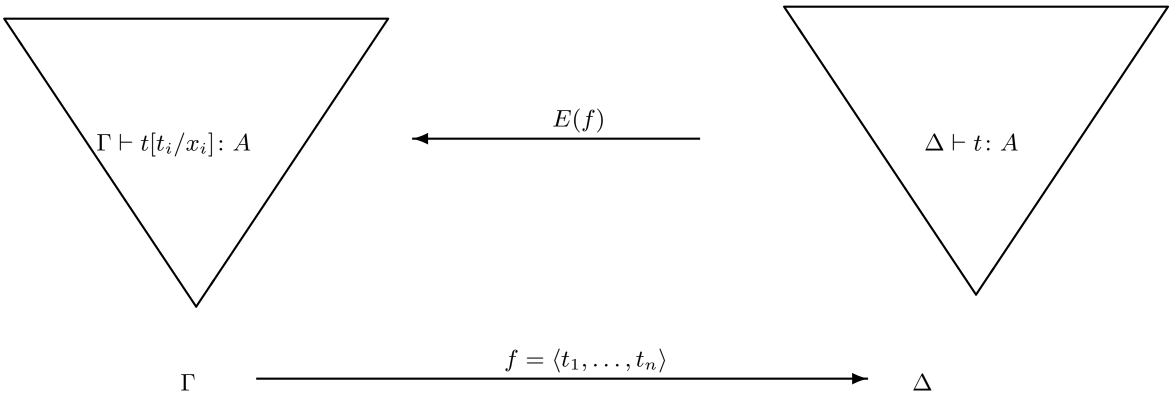

Figure 1: Modelling the simply-typed λ-calculus in a D-category

this does not bring in all the machinery required for the modality !.

The expert reader will note that the fibration modelling of intuitionistic logic is only necessary for dealing with predicates and/or dependent types; and this paper is only concerned with propositional intuitionistic (linear and non-linear) logic. However, fibration modelling does provide a means for adding linear type theory to intuitionistic type theory in the way we require, without a comonad or a monoidal adjunction for its effect and this is our main result here.

Also, the kind of modelling proposed in this paper can plausibly be extended to deal with dependent intuitionistic types.

5 Fibrational Models of Linear Logic

The fibration model interprets the dual context $\Gamma \mid \Delta$ as follows: the intuitionistic part of the context Γ is interpreted as an object of the base category while the whole context $\Gamma \mid \Delta$ gets interpreted as an object of the fibre over the interpretation of Γ. Thus the fibration model takes the view that terms are parametrised by the intuitionistic variables occurring in them, disregarding the linear ones. It does introduce an asymmetry between linear and intuitionistic variables, giving prominence to the intuitionistic variables, instead of the linear ones, but since change of fibre still corresponds to substitution of intuitionistic variables, as in the standard procedure, described, e.g. in Jacobs' book [Jac99], this is arguably a very natural construction. This was first described and presented at FOSSACS 2000 [MdPR00].

We show how to extend this categorical model of ILT using fibrations to a categorical model for DILL and LNLas well. This extension uses the fact that the modality ! can be characterised by a bijection between judgements

$$\frac{\Gamma \mid\, !A, \Delta \vdash M : B}{\overline{\Gamma, A \mid \Delta \vdash N : B}} \tag{1}$$

We show that this characterisation corresponds to a *symmetric monoidal adjunction* between the base category and the fibre over the empty context.

This adjunction therefore provides the means for relating our approach with double contexts to those type theories with only single contexts and a special type $!A$ (e.g. intuitionistic linear logic) for which weakening and contraction are available. Thus using fibrations we do not identify the interpretation of $\Gamma \mid \Delta \vdash M : A$ and $\_ \mid !\Gamma, \Delta \vdash M : A$, as we do when using the traditional, one-dimensional models.

A closer look at the isomorphism (1) indeed reveals that the possibility of shifting an intuitionistic assumption into a linear one via the modality implies that we can use one-dimensional semantics to interpret double contexts, i.e. we can use only the fibre over empty intuitionistic contexts and this is what happens when interpreting DILL and LNL via symmetric monoidal adjunctions (as in [BP97, Bar97, Ben95]). This way we can avoid fibrations. On the other hand the other direction of the isomorphism says that we can avoid the use of the modality by keeping the double context distinction. If we do not consider types $!!A$ and $A \multimap !B$, this makes it possible to avoid using the modality ! altogether. This is the choice that the calculus ILT makes.

We also prove a model-theoretic counterpart of this observation. We show that the *adjunction between the Grothendieck completion category and the base category* used to express the fibrational semantics of ILT is actually a symmetric monoidal adjunction and hence it turns out that this adjunction provides a DILL fibration model generated by an ILT-fibrational model. In other words, any ILT-fibrational model is already equipped with all what it is needed to interpret the modality, simply using the double contexts whose categorical counterpart is the Grothendieck completion. However, this modality is special in the sense that it is idempotent: $!!A$ is isomorphic to $!A$.

For applications in functional languages, types $!!A$ and $A \multimap !B$ are often unnecessary, as the important point is only whether an argument may be discarded or duplicated. We also show in the last section that DILL is a conservative extension of ILT (in other words, DILL just adds types $!!A$ and $A \multimap !B$ to ILT), thereby justifying the use of ILT in applications of linear logic to functional programming.

Let us now turn back to the categorical models for these calculi. As already mentioned, there are both fibrational and non-fibrational categorical models for DILL and LNL. We explained the fibrational categorical model for ILT, which can be stated rather succinctly via universal properties. The conservativity result also indicates that the definition of a non-fibrational model for ILT is problematic. Hence the canonical categorical model of ILT is the fibrational one. This paper uses a fibration to describe the relationship between the intuitionistic and linear parts of our model. This is orthogonal to the use fibrations for modelling polymorphic linear λ-calculi, as done by

Maneggia [Man04] and Birkedal et al. [MBP05].

We summarise the key points of the paper as follows:

- What is the categorical semantics of ILT? The fibration semantics is the most natural categorical semantics for ILT, as it models double contexts very well.
- How does this semantics extend to DILL and LNL? By adding a modality between the base category and the first fibre. This amounts to give a fibrational semantics to DILL and LNL. More interestingly, the fibrational semantics of DILL and LNL let us model categorically in terms of *an adjunction* the bijection in DILL between judgements The same holds for LNL with respect to the term constructors F and G.
- What is the difference in terms of expressive power between DILL and ILT? The answer is that DILL turns out to be a conservative extension of ILT, in other words DILL has more types but the set of closed terms of a given type which is common to DILL and ILT is the same.

The paper is organized as follows. After this long description of motivations, in the next section we describe the calculus ILT. In the following section we recall DILL and LNL. Then we define fibrational semantics for ILT in terms of IL-indexed categories and prove these sound and complete for ILT. We also show that ILT is the internal language of such categories. In the next section we provide fibrational semantics also for DILL and LNL. Of course these calculi have already been given traditional semantics [Bar97, BP97, Ben95], which we prove equivalent to ours, but the fibrational semantics seems more informative. Next we show that DILL is a conservative extension of ILT. We conclude with relating the fibrational and non-fibrational categorical semantics for all three calculi.

6 Intuitionistic and Linear Type Theory

The system of mixed intuitionistic and linear logic that we describe in this paper, called Intuitionistic and Linear Type Theory or ILT for short, borrows from Girard's Logic of Unity the elegant idea of separating assumptions into two classes: intuitionistic, which can be freely duplicated (shared) or discarded (ignored); and linear, which are constrained to be used exactly once. Syntactically, this strict separation is achieved by maintaining judgements with double-sided ("dual") contexts $\Gamma \mid \Delta \vdash A$, where, as a convention, Γ and Δ contain non-linear (intuitionistic) and linear assumptions, respectively. Another distinguishing feature of ILT is that it has both intuitionistic ($A \to B$) and linear implications ($A \multimap B$), as well multiplicative ($A \otimes B$) conjunctions with its unit (I), but no modality (or exponential) types $!A$. This system should not be confused with BI (the logic of bunched implications proposed by O'Hearn and Pym [OP99]). The multiplicative operators (tensor product and wand) are the same one as in linear logic, but product and function spaces are the additive version of the (multiplicative) tensor product and linear function spaces respectively.

The closest relative of the system ILT is Barber and Plotkin's DILL [BP97], which we will describe later. The key difference is that ILT has no terms (or morphisms) corresponding to the modality *per se*. This section describes the system ILT.

The set of types we shall work with is

$$A ::= A \multimap B \mid A \to B \mid A \otimes B \mid I$$

The syntax of preterms is defined inductively by

$$\begin{array}{rcl} M, N & ::= & a \mid x \mid \lambda a^A.M \mid \lambda x^A.M \mid M\,{}_I N \mid M_L N \\ & & \mid M \otimes N \mid \mathsf{let}\ M\ \mathsf{be}\ a \otimes b\ \mathsf{in}\ N \\ & & \mid \bullet \mid \mathsf{let}\ M\ \mathsf{be}\ \bullet\ \mathsf{in}\ N \end{array}$$

where a and x range over countable sets of linear and intuitionistic variables, respectively. This distinction of variables is not strictly necessary, but we adopt it here to aid legibility. Because the two let-expressions behave so similarly we sometimes write $\mathsf{let}\ M\ \mathsf{be}\ p\ \mathsf{in}\ N$ to cover both, where p (for a pattern) is either $a \otimes b$ or $\bullet$. ILT has typing judgements of the form $\Gamma|\Delta \vdash M \colon A$, where Γ and Δ are intuitionistic and linear contexts respectively, which are finite sets of typing declarations $x \colon A$ and $a \colon A$ for intuitionistic and linear variables respectively, with no variable repeated. The typing rules for ILT are given in Table 1. Note that in binary rules intuitionistic contexts are composed additively, whereas linear contexts are composed multiplicatively. Hence contraction for intuitionistic contexts is admissible, whereas this is not true for linear contexts. Also note the extra contexts Γ and Δ in the rules for intuitionistic and linear variables. These ensure that weakening for intuitionistic contexts is admissible, too.

We have three kinds of equations: β and η-equations and commuting conversions. For the presentation of these equations we use contexts-with-holes, written $C[\_]$. They are given by the grammar

$$\begin{array}{rcl} C[\_] & ::= & \_ \mid \lambda a^A.C[\_] \mid \lambda x^A.C[\_] \mid C[\_]_I M \mid M_I C[\_] \mid C[\_]_L M \mid M_L C[\_] \\ & & \mid C[\_] \otimes M \mid M \otimes C[\_] \mid \mathsf{let}\ C[\_]\ \mathsf{be}\ p\ \mathsf{in}\ N \mid \mathsf{let}\ M\ \mathsf{be}\ p\ \mathsf{in}\ C[\_] \end{array}$$

Note that this definition implies that there is exactly one occurrence of the symbol $\_$ in a context-with-hole $C[\_]$. The term $C[M]$ denotes the replacement of $\_$ in $C[\_]$ by M with the possible capture of free variables. This capture is the difference between the replacement of $\_$ and substitution for a free variable: if $C[\_]$ is the context-with-hole $(\lambda a^A.\_)$, then $C[a] = \lambda a^A.a$, whereas $(\lambda a^A.b)[a/b] = \lambda c^A.a$. The equations for ILT are given in Table 2.

$$\Gamma \mid a : A \vdash a : A \qquad \Gamma, x : A, \Delta \mid \_ \vdash x : A$$

$$\frac{\Gamma \mid \Delta, a : A \vdash M : B}{\Gamma \mid \Delta \vdash \lambda a^A.M : A \multimap B} \qquad \frac{\Gamma \mid \Delta_1 \vdash M : A \multimap B \quad \Gamma \mid \Delta_2 \vdash N : A}{\Gamma \mid \Delta \vdash M_L N : B}$$

$$\frac{\Gamma, x : A \mid \Delta \vdash M : B}{\Gamma \mid \Delta \vdash \lambda x^A.M : A \to B} \qquad \frac{\Gamma \mid \Delta \vdash M : A \to B \quad \Gamma \mid \_ \vdash N : A}{\Gamma \mid \Delta \vdash M_I N : B}$$

$$\frac{\Gamma \mid \Delta_1 \vdash M : A \quad \Gamma \mid \Delta_2 \vdash N : B}{\Gamma \mid \Delta \vdash M \otimes N : A \otimes B}$$

$$\frac{\Gamma \mid \Delta_1 \vdash M : A \otimes B \quad \Gamma \mid a : A, b : B, \Delta_2 \vdash N : C}{\Gamma \mid \Delta \vdash \mathsf{let}\ M\ \mathsf{be}\ a \otimes b\ \mathsf{in}\ N : B}$$

$$\frac{}{\Gamma \mid \_ \vdash \bullet : I} \qquad \frac{\Gamma \mid \Delta_1 \vdash M : I \quad \Gamma \mid \Delta_2 \vdash N : C}{\Gamma \mid \Delta \vdash \mathsf{let}\ M\ \mathsf{be}\ \bullet\ \mathsf{in}\ N : C}$$

Where applicable Δ_1, Δ_2 are disjoint and Δ is a permutation of Δ_1, Δ_2.

Table 1: The typing rules of ILT

Note that in ILT linear variables can move across the divisor of the context as expressed in the following lemma.

Lemma 1 *For every ILT derivable judgement* $\Gamma \mid a_1 : A_1, \ldots, a_n : A_n, \Sigma \vdash M : B$ *we can derive* $\Gamma, x_1 : A_1, \ldots, x_n : A_n \mid \Sigma \vdash M[x_1/a_1, \ldots x_n/a_n] : B$.

Although in this paper we are presenting only equational theories, ILT gives rise in a natural way to a reduction theory, namely by orienting all β-equations from left to right and by adding commuting conversions rules for the tensor. To avoid failure of strong normalisation, only those commuting conversions which shift let-expressions towards the top of the syntax tree are added as reduction rules. In this way one obtains a strongly normalising and confluent reduction theory. For details about this way of turning an equational theory into a reduction theory see [GdPR00].

So far, we have considered only the set of types and terms required to express the logical constructors. However, to make this calculus interesting one needs to add additional (non-logical) types and terms and equations between terms (for example the type of natural numbers and terms for successor, addition etc.). We call the calculus with such additional types and terms an ILT-theory. The syntactic properties of the ILT-calculus can be extended to ILT-theories as well under reasonable assumptions.

β-equations:

$$(\lambda x^A.M)_I N = M[N/x] \qquad (\lambda a^A.M)_L N = M[N/a]$$
$$\mathsf{let}\ M{\otimes}N\ \mathsf{be}\ a{\otimes}b\ \mathsf{in}\ R = R[M/a, N/b] \qquad \mathsf{let}\ \bullet\ \mathsf{be}\ \bullet\ \mathsf{in}\ M = M$$

η-equations:

$$\lambda a\cdot M_L a = M \qquad \lambda x^A.M_I x = M \ \text{ if } x \notin FV(M)$$
$$\mathsf{let}\ M\ \mathsf{be}\ a{\otimes}b\ \mathsf{in}\ a{\otimes}b = M \qquad \mathsf{let}\ M\ \mathsf{be}\ \bullet\ \mathsf{in}\ \bullet = M$$

Commuting conversions:

$$\begin{array}{rcl}
\mathsf{let}\ M\ \mathsf{be}\ *\ \mathsf{in}\ C[N] & = & C[\mathsf{let}\ M\ \mathsf{be}\ *\ \mathsf{in}\ N] \\
\mathsf{let}\ M\ \mathsf{be}\ a{\otimes}b\ \mathsf{in}\ C[N] & = & C[\mathsf{let}\ M\ \mathsf{be}\ a{\otimes}b\ \mathsf{in}\ N]
\end{array}$$

Table 2: The equations of ILT

7 More linear type theories: DILL and LNL

In this section we discuss two other linear type theories, which also have dual or double contexts. These, unlike ILT, have categorical semantics already given in the literature. The fibrational semantics we present in the next section for these type theories is perhaps harder to grasp than the one originally given, but as we shall argue, these new fibrational semantics are more faithful to the calculi: they do not make the simplifying assumption that the double context $\Gamma \mid \Delta$ can (and for simpler semantics has to) be transformed into the simpler context $\_ \mid !\Gamma, \Delta$.

7.1 Dual Intuitionistic and Linear Logic DILL

The type theory DILL was first considered by Plotkin and his student Barber [BP97, Bar97] more or less at the same time as LNL was investigated by Benton. DILL has judgements of the same form and with the same components as ILT, namely $\Gamma \mid \Delta \vdash M\colon A$, where Γ is an intuitionistic context, Δ is a linear context, M a term and A a type. The important difference is that instead of two implications as in ILT, DILL has only the linear implication $\multimap$ and the modal type $!A$ with appropriate term constructors.

$$A ::= I \mid A \otimes A \mid A \multimap A \mid !A$$

Preterms are similar to the ones in ILT for the common types

$$\begin{array}{rcl}
M, N & ::= & a \mid x \mid \lambda a^A.M \mid M_L N \\
 & & \mid M \otimes N \mid \mathsf{let}\ M\ \mathsf{be}\ a \otimes b\ \mathsf{in}\ N \\
 & & \mid \bullet \mid \mathsf{let}\ M\ \mathsf{be}\ \bullet\ \mathsf{in}\ N
\end{array}$$

plus term constructors for !:

$$M, N \quad ::= \quad !M \mid \mathsf{let}\ M\ \mathsf{be}\ !x\ \mathsf{in}\ N$$

The typing rules of DILL are the ones from ILT, again excluding intuitionistic implications plus the following rules relating to the modality (or exponential) "!":

$$\frac{\Gamma \mid \_ \vdash M : A}{\Gamma \mid \_ \vdash !M : !A} \qquad \frac{\Gamma \mid \Delta_1 \vdash M : !A \quad \Gamma, x \colon A \mid \Delta_2 \vdash N \colon B}{\Gamma \mid \Delta \vdash \mathsf{let}\ M\ \mathsf{be}\ !x\ \mathsf{in}\ N : B}$$

Note that Δ_1, Δ_2 are disjoint and Δ is a permutation of Δ_1, Δ_2. Also note that the axiom $\Gamma, x \colon A \_ \vdash x \colon A$ corresponds to the *dereliction* rule for the modality !, i.e. it allows for movement of assumptions from the intuitionistic part to the linear part of the context.

The equations for DILL are the ones for ILT (minus the ones relating to intuitionistic implications) plus the following β and generalised η-rules for the modality:

$$\mathsf{let}\ !M\ \mathsf{be}\ !x\ \mathsf{in}\ N = N[M/x] \qquad \mathsf{let}\ M\ \mathsf{be}\ !x\ \mathsf{in}\ N[!x/a] = N[M/a] \colon B$$

In the same way as ILT, DILL has a natural reading as a reduction theory, which satisfies subject reduction, strong normalization and confluence.

The intuitionistic part of the double context can be represented by a linear context with !-types. The details are as follows:

Proposition 2 (Barber) *In DILL there is a an isomorphism* Φ *between the set of all terms* M *such that* $\Gamma \mid a \colon !A, \Delta \vdash M \colon B$ *and all terms* N *such that* $\Gamma, x \colon A \mid \Delta \vdash N \colon B$.

Proof. Define Φ by transforming a term M such that $\Gamma \mid a \colon !A, \Delta \vdash M \colon B$ to the term $M[!x/a]$, and the inverse Φ^{-1} by $\Phi^{-1}(N) = \mathsf{let}\ a\ \mathsf{be}\ !x\ \mathsf{in}\ N$. The verification that these two maps are indeed inverse to each other is routine. $\square$

7.2 Linear-non-Linear logic LNL

Benton's type theory LNL [Ben95, Ben94] attaches equal prominence to both linear and intuitionistic logic, instead of giving linear logic the status of the basic logic.

The type theory LNL is split into a linear part and an intuitionistic part, each one with a different kind of judgement. The intuitionistic part is the simply typed λ-calculus. The type constructor !, which regulates when weakening and contraction are applicable, is replaced by an appropriate relation between the linear and the intuitionistic parts of the type theory. This special relation between the intuitionistic and the linear parts of the type theory is given by two type constructors F and G which map the intuitionistic type theory into the linear one and vice-versa, and terms which describe how these type constructors interact. This type theory was derived from the categorical semantics, and we hence we will describe the intuition behind F and G when we discuss the categorical semantics. Originally, the type theory LNL had products (or additive conjunctions) on its intuitionistic part but for this paper we call LNL the theory without the additive conjunction.

Types (linear and non-linear) are as follows:

$$A \ ::= \ I \mid A \otimes A \mid A \multimap A \mid F(\sigma) \qquad \sigma ::= G(A) \mid \sigma \to \sigma$$

We write A, B, C etc. for linear types and σ, τ etc. for non-linear types. Raw terms (linear and non-linear) of the LNL-type theory are given by:

$$\begin{array}{rcl} M & ::= & a \mid \bullet \mid \mathsf{let}\ M\ \mathsf{be}\ \bullet\ \mathsf{in}\ M \mid \lambda a^A.M \mid M_L M \mid \\ & & M \otimes M \mid \mathsf{let}\ M\ \mathsf{be}\ a \otimes b\ \mathsf{in}\ M \mid F(t) \mid \mathsf{let}\ M\ \mathsf{be}\ F(x)\ \mathsf{in}\ M \mid \mathsf{derelict}(t) \\ t & ::= & x \mid \lambda x^\sigma.t \mid t_I t \mid G(M) \end{array}$$

We write M, N etc. for linear terms and t, s etc. for non-linear terms. We have two kinds of judgements, $\Gamma|\Delta \vdash_{\mathcal{L}} M \colon A$ and $\Gamma \vdash_{\mathcal{C}} t \colon \sigma$. The first one captures the *linear* part of the type theory (hence the subscript $\mathcal{L}$), whereas the second one captures the intuitionistic part (here $\mathcal{C}$ stands for *cartesian*). The linear inference rules contain all the inference rules for ILT for linear variables and the unit type, tensor products and linear function spaces. The intuitionistic inference rules contain all the rules for the simply-typed λ-calculus. In addition we have the following rules modelling the interaction between the linear and the intuitionistic parts:

$$\frac{\Gamma \vdash_{\mathcal{C}} t \colon \sigma}{\Gamma|\_ \vdash_{\mathcal{L}} F(t) \colon F(\sigma)} \qquad \frac{\Gamma|\Delta_1 \vdash_{\mathcal{L}} M \colon F(\sigma) \qquad \Gamma, x \colon \sigma|\Delta_2 \vdash_{\mathcal{L}} N \colon B}{\Gamma|\Delta \vdash_{\mathcal{L}} \mathsf{let}\ M\ \mathsf{be}\ F(x)\ \mathsf{in}\ N \colon B}$$

$$\frac{\Gamma|\_ \vdash_{\mathcal{L}} M \colon A}{\Gamma \vdash_{\mathcal{C}} G(M) \colon G(A)} \qquad \frac{\Gamma \vdash_{\mathcal{C}} t \colon G(A)}{\Gamma|\_ \vdash_{\mathcal{L}} \mathsf{derelict}(t) \colon A}$$

The equations for intuitionistic terms, derived using the judgement $\vdash_{\mathcal{C}}$, are those for the simply typed lambda calculus. The equations for the linear terms, derived using the judgement $\vdash_{\mathcal{L}}$, are the ones from ILT plus the following two β and η-equations:

$$\frac{\Gamma \vdash_{\mathcal{C}} t\colon \sigma \qquad \Gamma, x\colon \sigma|\Delta \vdash_{\mathcal{L}} M\colon A}{\Gamma|\Delta \vdash_{\mathcal{L}} \mathsf{let}\ F(t)\ \mathsf{be}\ F(x)\ \mathsf{in}\ M = M[t/x]\colon A}\ (F\beta)$$

$$\frac{\Gamma|\Delta_1 \vdash_{\mathcal{L}} N\colon F(\sigma) \qquad \Gamma|a\colon F(\sigma), \Delta_2 \vdash_{\mathcal{L}} M\colon A}{\Gamma|\Delta_1, \Delta_2 \vdash_{\mathcal{L}} \mathsf{let}\ N\ \mathsf{be}\ F(x)\ \mathsf{in}\ M[F(x)/a] = M[N/a]\colon A}\ (F\eta)$$

$$\frac{\Gamma|\_ \vdash_{\mathcal{L}} M\colon A}{\Gamma|\_ \vdash_{\mathcal{L}} \mathsf{derelict}(G(M)) = M\colon A}\ (G\beta)$$

$$\frac{\Gamma \vdash_{\mathcal{C}} t\colon G(A)}{\Gamma \vdash_{\mathcal{C}} G(\mathsf{derelict}(t)) = t\colon G(A)}\ (G\eta)$$

Using the same arguments as for tensor products, one can show that the generalised η-rule for F is equivalent to the standard η-rule ($\mathsf{let}\ M\ \mathsf{be}\ F(x)\ \mathsf{in}\ F(x) = M$) plus commuting conversions.

Similarly to DILL we can show that intuitionistic contexts can be replaced by linear ones with F-types:

Proposition 3 *In LNL there is a an isomorphism between the set of all terms M such that $\Gamma \mid a\colon F(\sigma), \Delta \vdash M\colon B$ and all terms N such that $\Gamma, x\colon \sigma \mid \Delta \vdash N\colon B$.*

7.3 Relating ILT and DILL

We can translate ILT into DILL by translating the intuitionistic implication as $!A \multimap B$ and the terms accordingly, and leaving all other terms unaffected. This translation was described already in a different context in Girard's original paper on linear logic [Gir87b]. The precise definition is as follows:

Definition 4 *By induction over the structure of ILT-types and ILT-preterms we define a translation from ILT to DILL called $(\_)^{DILL}$ as follows:*

(i) *$(\_)^{DILL}$ maps all type constructors of ILT except the intuitionistic function space into the same type constructor in DILL;*

(ii) $(A \to B)^{DILL} = !(A)^{DILL} \multimap (B)^{DILL}$;

(iii) *$(\_)^{DILL}$ maps all term constructors except λ-abstraction over intuitionistic variables and intuitionistic application to the same term constructor in DILL;*

(iv)

$$\begin{array}{rcl} (\lambda x^A.M)^{DILL} & = & \lambda a^{!(A)^{DILL}}.\mathsf{let}\ a\ \mathsf{be}\ !x\ \mathsf{in}\ (M)^{DILL} \\ (M_I N)^{DILL} & = & (M)^{DILL}_L(!(N)^{DILL}) \end{array}$$

The translation $(\_)^{DILL}$ is extended to contexts componentwise;

This translation respects judgements:

Proposition 5 *(i) Assume* $\Gamma|\Delta \vdash M\colon A$. *Then also* $(\Gamma)^{DILL}|(\Delta)^{DILL} \vdash (M)^{DILL}\colon (A)^{DILL}$.

(ii) Assume $\Gamma|\Delta \vdash M = N\colon A$. *Then also* $(\Gamma)^{DILL}|(\Delta)^{DILL} \vdash (M)^{DILL} = (N)^{DILL}\colon (A)^{DILL}$.

Hence DILL considered as a type theory is at least as powerful as ILT. We will see later that DILL is in fact a conservative extension of ILT.

We will clarify the precise relationship when we have described the semantics for both type theories in detail.

8 Categorical semantics of ILT

The basis for our categorical model of ILT is Ehrhard's notion of a D-category for modelling dependent types [Ehr88], which goes back to Lawvere's idea of hyperdoctrines satisfying the comprehension axiom [Law70]. Hyperdoctrines model many-sorted predicate logic, where predicates are indexed over sorts or sets. More precisely, a hyperdoctrine is an indexed category where the fibres are partial orders. The base category models the terms and sorts, and the fibre over an object corresponding to a list of sorts models the predicates indexed by these sorts. A suitable adjunction allows the interpretation of the comprehension axiom, that is the creation of a subset defined by a predicate indexed over a set. This idea has been adopted also to model other notions of dependency in type theory, *e.g.*,dependent types where types and terms are indexed by contexts, ie lists of types. For modelling these one considers indexed categories where the fibres model the types and terms and the base category models the contexts. [Ehr88, Jac99, Str91]. In this case the adjunction which corresponds to the comprehension axiom models the creation of contexts by adding more types (which depend on previously declared contexts).

We illustrate these concepts by describing D-categories [Ehr88] simplified to give models of the simply-typed λ-calculus. A D-category is a split indexed category $E\colon \mathcal{B}^{op}\to\mathbf{Cat}$[1] where the base category $\mathcal{B}$ models contexts and the fibre over an object Γ models terms whose free variables are contained in the context modelled in Γ. We require both $\mathcal{B}$ and each fibre $E(\Gamma)$, for Γ in $\mathcal{B}$, to have *a terminal object*, respectively $\top_{\mathcal{B}}$ and $\top_{\Gamma}$. We also require that for every f morphism in the base category $E(f)$ *preserves the terminal object*. The fibration associated to this indexed category is the projecting functor $p : Gr(E)\to\mathcal{B}$, where $Gr(E)$ is the Grothendieck completion (see also page 107 of [Jac99]). We recall that the objects of the Grothendieck

[1]Note that from now on when we refer to indexed categories we mean split indexed categories, i.e. the pseudofunctor towards **Cat** is actually a functor.

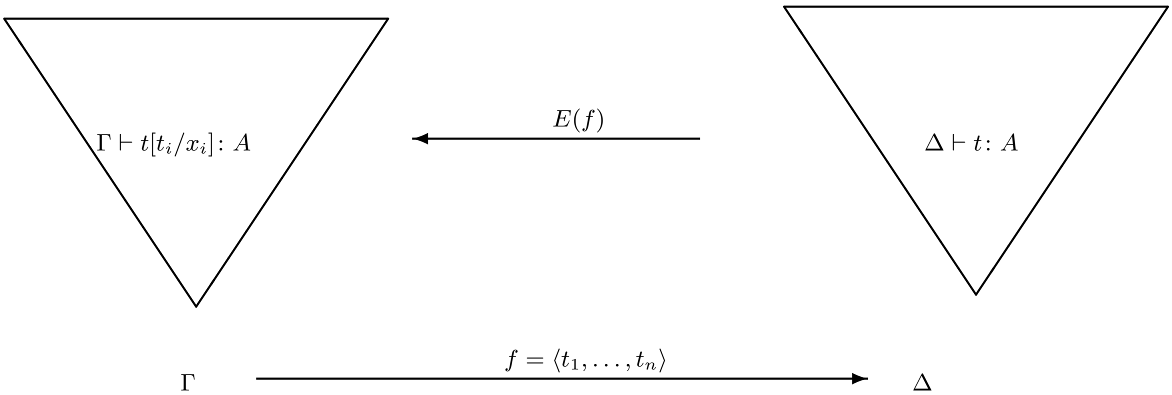

Figure 2: Modelling the simply-typed λ-calculus in a D-category

completion of E are the pairs (Γ, A) where Γ is an object of $\mathcal{B}$ and A is an object of $E(\Gamma)$. The morphisms of $Gr(E)$ between (Γ, A) and (Δ, C) are pairs (f, h) where $f : \Gamma \to \Delta$ is a morphism in $\mathcal{B}$ and $h : A \to E(f)(C)$ is a morphism in $E(\Gamma)$. For every object Γ in $\mathcal{B}$ the category $E(\Gamma)$ is called the fibre of E over the object Γ.

The key construction of a D-category to interpret contexts and substitutions is the requirement called the "comprehension property" i.e. the requirement that the terminal object functor $\mathcal{T}\colon \mathcal{B} \to Gr(E)$ has got a *right adjoint* $P\colon Gr(E) \to \mathcal{B}$. Recall that the functor $\mathcal{T}$ is defined as follows: for every object Γ in the base category $\mathcal{B}$, $\mathcal{T}(\Gamma) = (\Gamma, \top_\Gamma)$ and for every morphism f, $\mathcal{T}(f) = (f, \mathsf{Id})$. Actually $\mathcal{T}$ is an embedding functor of the base category $\mathcal{B}$ into the fibres of E. The right adjoint to $\mathcal{T}$ ensures firstly that contexts are lists of objects of the base category over the empty type. Secondly, the adjunction also ensures that a morphism in the fibre corresponds to a morphism in the base category and this allows to model substitution by the re-indexing functor. This intuition is captured in Figure 2. The idea for the model of ILT is to modify this setting to capture the separation between intuitionistic and linear variables in ILT (with their corresponding substitutions) and simultaneously to model the two identity axioms, i.e. the assumptions of intuitionistic and linear variables, acting on the same types. The base category $\mathcal{B}$ models only the intuitionistic contexts of ILT, *i.e.*, objects in $\mathcal{B}$ model contexts $(\Gamma \mid \_)$. Each fibre over an object in $\mathcal{B}$ modelling a context $\Gamma \mid \_$ models terms $\Gamma \mid \Delta \vdash M\colon A$ for any context Δ. We require a terminal object in $\mathcal{B}$. The fibres are now symmetric monoidal closed categories (SMC-categories) and model the linear constructions of ILT. The functors between the fibres have to preserve the SMC structure.

We replace the right adjoint to the terminal object functor $\mathcal{T}$ by a right adjoint P to the unit functor $U\colon \mathcal{B} \to Gr(E)$, assigning to each object Γ in $\mathcal{B}$ the object (Γ, I). This right-adjoint P is the comprehension functor. In this way we obtain that morphisms in the base correspond to morphisms with domain I in the fibre, *i.e.*, terms with no free linear variables.

Now, we can model substitution for intuitionistic variables by reindexing along morphisms in the base as usual: this adjunction $U \dashv P$ enforces the restriction that only terms with no free linear variables can be substituted for intuitionistic variables.

The structure discussed so far is common to all type theories discussed in this paper. It consists of ILT except the intuitionistic function spaces. It has a categorical semantics described as follows:

Definition 6 *Let $\mathcal{B}$ be a category with a terminal object $\top_{\mathcal{B}}$. A* simple linear indexed category *is a functor $E\colon \mathcal{B}^{op} \to \mathbf{Cat}$ such that the following conditions are satisfied. (Note that we write f^* for the application of the functor E to f, for any morphism f in $\mathcal{B}$.)*

(i) Fibre structure. *$E(\Gamma)$ is a symmetric monoidal closed category, i.e. a SMC category, for each object Γ of $\mathcal{B}$. Moreover for each morphism f in $\mathcal{B}$, the functor f^* preserves this SMC structure on the nose, i.e. it is a SMC functor.*

(ii) Comprehension. *For each object Γ of $\mathcal{B}$ the functor $U\colon \mathcal{B} \to Gr(E)$ given by $U(\Gamma) = (\Gamma, I)$ and $U(f) = (f, \mathsf{Id})$, has a right adjoint $P\colon Gr(E) \to \mathcal{B}$. The object $P(\Gamma, A)$ is abbreviated $\Gamma.A$ in the sequel and the morphism $P(f, h)$ is written $f.h$.*
Furthermore, $(\mathsf{Fst}_A, \mathsf{Snd}_A)\colon (\Gamma.A, I) \to (\Gamma, A)$ denotes the counit of this adjunction. [2] *The natural isomorphism between $\mathsf{Hom}_{Gr(E)}((-, I), (-, A))$ and $\mathsf{Hom}_{\mathcal{B}}(-, -.A)$ is denoted by $[-,-]$ (and its inverse by $[-,-]^{-1}$)*

(iii) Non-dependency. *For every object $\Delta \in Ob(\mathcal{B})$, meaning with $\iota_\Delta : \Delta \to \top_{\mathcal{B}}$ the unique map towards the terminal object $\top_{\mathcal{B}}$, the functor $E_\top(\iota_\Delta) : E_\top(\top) \to E(\Delta)$ is bijective on the objects.*

We need to add further structure to model the intuitionistic function spaces of ILT, namely the right adjoint to weakening.

Definition 7 *An* ILT-indexed category *is a $E\colon \mathcal{B}^{op} \to \mathbf{Cat}$ is a simple linear indexed category satisfying:*

(i) Right adjoint to weakening. *For every object Γ of $\mathcal{B}$ and A of $E(\Gamma)$, the functor $\mathsf{Fst}^*_A\colon E(\Gamma) \to E(\Gamma.A)$ has a right adjoint $\Pi_A\colon E(\Gamma.A) \to E(\Gamma)$. We will write in the sequel Cur_I for the natural isomorphism between $\mathrm{Hom}_{E(\Gamma.A)}(\mathsf{Fst}^*_A(B), C)$ and $\mathrm{Hom}_{E(\Gamma)}(B, \Pi_A(C))$ and App_I for its counit.*

(ii) Beck-Chevalley condition. *The Beck-Chevalley-condition for the adjunction $\mathsf{Fst}^*_A \vdash \Pi_A$ is satisfied in the strict sense, i.e. the equations*

$$f^* \cdot \Pi_A = \Pi_A \cdot (f.\mathsf{Id})^* \qquad f^*(\mathsf{Cur}_I(t)) = \mathsf{Cur}_I((f.\mathsf{Id})^*(t))$$

hold for every $f\colon \Delta \to \Gamma$, $A \in ObE(\Gamma)$, $B \in ObE(\Gamma)$ and $t \in \mathrm{Hom}_{E(\Gamma.A)}(\mathsf{Fst}^*_A(B), C)$.

[2] We sometimes omit the subscripts.

(iii) $\mathsf{Contextual\ base}$. *The objects of the category* $\mathcal{B}$ *are either* $\top_{\mathcal{B}}$ *or* $\top_{\mathcal{B}}.A_1.A_2.\ \ldots\ .A_n$ *where each* A_i *from* $i = 1, \ldots, n$ *is an object of* $E_{\top_{\mathcal{B}}}$.

Note that we mean preservation on the nose when we say that a functor preserves some categorical structure, which in turn is considered to be fixed.

The interpretation of intuitionistic contexts requires products in the base category $\mathcal{B}$ of an ILT-indexed category. Although not required by definition, it is easy to show that in fact the category $\mathcal{B}$ is always cartesian:

Proposition 8 *Given an* ILT-indexed category $E\colon \mathcal{B}^{op} \to \mathbf{Cat}$ *the base category* $\mathcal{B}$ *is cartesian.*

Proof. The product of $\top_{\mathcal{B}}.A$ and $\top_{\mathcal{B}}.C$ is $\top_{\mathcal{B}}.A.C$ where we simply write C instead of $E(\mathsf{Fst}_A)(C)$. From this it is then clear how the product between any two generic objects of $\mathcal{B}$ can be defined. The first projection is defined as $p_1 = \mathsf{Fst}_C : \top_{\mathcal{B}}.A.C \to \top_{\mathcal{B}}.A$ and the second projection as

$$p_2 = [\mathsf{Fst}_A \cdot \mathsf{Fst}_C, \mathsf{Snd}_C] : \top_{\mathcal{B}}.A.C \to \top_{\mathcal{B}}.C$$

□

Next, we define the interpretation of the ILT-calculus in an ILT-indexed category.

Definition 9 *Given any ILT-indexed category* $E\colon \mathcal{B}^{op} \to \mathbf{Cat}$*, an ILT-theory, a map* $[\![\_]\!]$ *from ground types to objects in* $E(\top_{\mathcal{B}})$ *and a map from ground terms* f *such if that* $x_1\colon G_1, \ldots, x_n\colon G_n | a_1\colon G'_1, \ldots, a_m\colon G'_m \vdash f\colon G$ *then* $[\![f]\!]$ *is a map from* $[\![G'_1]\!] \otimes \cdots [\![G'_m]\!]$ *to* $[\![G]\!]$ *in* $E([\![G_1]\!] \times \cdots \times [\![G_n]\!])$*. We extend this map to map* $[\![\_]\!]$ *from types to objects in* $E(\top_{\mathcal{B}})$*, from intuitionistic contexts* Γ *to objects in* $\mathcal{B}$*, from linear contexts* Δ *to objects in* $E(\top_{\mathcal{B}})$*, from double contexts to objects of suitable fibres and from terms* $\Gamma \mid \Delta \vdash M\colon A$ *to morphisms* $[\![M]\!]\colon [\![\Delta]\!] \to [\![A]\!]$ *in* $E([\![\Gamma]\!])$ *by induction over the structure:*

(i) *On intuitionistic and linear contexts respectively:*

$$[\![\_]\!] = \top_{\top_{\mathcal{B}}} \quad [\![\Gamma, x\colon A]\!] = [\![\Gamma]\!].[\![A]\!] \qquad\qquad [\![\_]\!] = I \quad [\![\Delta, a\colon A]\!] = [\![\Delta]\!] \otimes [\![A]\!]$$

where I *is the tensor-unit in the category* $E(\top_{\mathcal{B}})$ *and also* $[\![\Delta]\!]$ *and* $[\![A]\!]$ *are objects of* $E(\top_{\mathcal{B}})$*.*

On double contexts: $[\![\Gamma \mid \Delta]\!] = \iota^*_{[\![\Gamma]\!]}([\![\Delta]\!])$ *because* $[\![\Delta]\!]$ *being a linear context is an object of* $E(\top_{\mathcal{B}})$*.*

(ii) *On types:*

$$[\![A \to B]\!] = \Pi_{[\![A]\!]}([\![B]\!]) \quad [\![A \multimap B]\!] = [\![A]\!] \multimap [\![B]\!] \quad [\![A \otimes B]\!] = [\![A]\!] \otimes [\![B]\!] \quad [\![I]\!] = I$$

(iii) On terms (assuming that $\Gamma = x_1 \colon A_1, \ldots, x_n \colon A_n$):

$$[\![\Gamma, x\colon A, y\colon C \mid \_ \vdash x\colon A]\!] = \mathsf{Fst}_{[\![\Gamma,x:A,y:C]\!]}{}^*(\mathsf{Snd}_{[\![\Gamma,x:A]\!]})$$

$$[\![\Gamma \mid a\colon A \vdash a\colon A]\!] = \mathsf{Id}$$

$$\frac{[\![\Gamma, x^A \mid \Delta \vdash M\colon B]\!] = t}{[\![\Gamma \mid \Delta \vdash \lambda x^A.M\colon A \to B]\!] = \mathsf{Cur}_I(t)}$$

$$\frac{[\![\Gamma \mid \Delta \vdash M\colon A \to B]\!] = t \qquad [\![\Gamma \mid \_ \vdash N\colon A]\!] = s}{[\![\Gamma \mid \Delta \vdash M_I N\colon B]\!] = [\mathsf{Id}, s]^*(\mathsf{App}_I \cdot \mathsf{Fst}^*(t))}$$

$$\frac{[\![\Gamma \mid \Delta_1 \vdash M\colon A]\!] = t \qquad [\![\Gamma \mid \Delta_2 \vdash N\colon B]\!] = s}{[\![\Gamma \mid \Delta \vdash M \otimes N\colon A \otimes B]\!] = (t \otimes s) \cdot \pi}$$

$$\frac{[\![\Gamma \mid \Delta_1 \vdash M\colon A \otimes B]\!] = m \qquad [\![\Gamma \mid \Delta_2, a\colon A, b\colon B \vdash N\colon C]\!] = n}{[\![\Gamma \mid \Delta \vdash \mathsf{let}\ M\ \mathsf{be}\ a \otimes b\ \mathsf{in}\ N\colon C]\!] = n \cdot (\mathsf{Id} \otimes m) \cdot \pi}$$

$$[\![\Gamma \mid \_ \vdash \bullet : I]\!] = \mathsf{Id} \qquad \frac{[\![\Gamma \mid \Delta_1 \vdash M\colon I]\!] = m \qquad [\![\Gamma \mid \Delta_2 \vdash N\colon C]\!] = n}{[\![\Gamma \mid \Delta \vdash \mathsf{let}\ M\ \mathsf{be}\ \bullet\ \mathsf{in}\ N\colon C]\!] = n \cdot {r_I}^{-1} \cdot (\mathsf{Id} \otimes m) \cdot \pi}$$

$$\frac{[\![\Gamma \mid \Delta, a\colon A \vdash M\colon B]\!] = t}{[\![\Gamma \mid \Delta \vdash \lambda a^A.M\colon A \multimap B]\!] = \mathsf{Cur}_L(t)}$$

$$\frac{[\![\Gamma \mid \Delta_1 \vdash M\colon A \multimap B]\!] = t \qquad [\![\Gamma \mid \Delta_2 \vdash N\colon A]\!] = s}{[\![\Gamma \mid \Delta \vdash M_L N\colon B]\!] = \mathsf{App}_L \cdot (t \otimes s) \cdot \pi}$$

where π is the canonical morphism from $[\![\Delta]\!]$ to $[\![\Delta_1]\!] \otimes [\![\Delta_2]\!]$, r_I the suitable part of the isomorphism between $[\![\Delta_2]\!] \otimes I$ and $[\![\Delta_2]\!]$ with inverse ${r_I}^{-1}$, Cur_L the corresponding natural transformation for the adjunction between $(-) \otimes [\![A]\!]$ and $[\![A]\!] \multimap (-)$ in $E([\![\Gamma]\!])$ and App_L the related counit.

Next, we turn to the soundness of this categorical semantics. As always, the key lemmata concern substitution. In particular they are needed to prove the validity of introduction and elimination rules regarding intuitionistic implication and of all the conversion rules involving substitution. As we have two kinds of substitution, we have to show two substitution lemmata, namely for substitution of intuitionistic and linear variables.

Lemma 10 *(i) Assume $[\![\Gamma, x\colon A \mid \Delta \vdash M\colon B]\!] = t$ and $[\![\Gamma \mid \_ \vdash N\colon A]\!] = s$. Then $[\![\Gamma \mid \Delta \vdash M[N/x]\colon B]\!] = [\mathsf{Id}, s]^*(t)$.*

(ii) Assume $[\![\Gamma \mid \Delta_1, a\colon A \vdash M\colon B]\!] = t$ and $[\![\Gamma \mid \Delta_2 \vdash N\colon A]\!] = s$. Then $[\![\Gamma \mid \Delta \vdash M[N/a]\colon B]\!] = t \cdot (\mathsf{Id} \otimes s) \cdot \pi$, where π is the canonical morphism from $[\![\Delta]\!]$ to $[\![\Delta_1 \otimes \Delta_2]\!]$.

Proof. Induction over the structure of M. □

The soundness proof is now routine.

Theorem 11 *Given an IL- indexed category $E\colon \mathcal{B}^{op} \to$ **Cat** under the above interpretation $[\![]\!]$ the following facts hold.*

(i) Assume $\Gamma \mid \Delta \vdash M\colon A$. Then $[\![\Gamma \mid \Delta \vdash M\colon A]\!]$ is a morphism from $[\![\Delta]\!]$ to $[\![A]\!]$ in $E([\![\Gamma]\!])$;

(ii) Assume $\Gamma \mid \Delta \vdash M = N\colon A$. Then $[\![\Gamma \mid \Delta \vdash M\colon A]\!] = [\![\Gamma \mid \Delta \vdash N\colon A]\!]$.

Next, we want to prove the completeness theorem and as usual its proof is based on the construction of a term model out of ILT which we are going to describe. For a start, recall that in order to prove that two functors $U : \mathcal{B} \to Gr(E)$ and $P : Gr(E) \to \mathcal{B}$ define an adjunction $U \dashv G$, it suffices to give two data: firstly, a natural transformation $\alpha_D : \mathsf{Hom}(U(-), D) \to \mathsf{Hom}(-, P(D))$ for each object D in $Gr(E)$, and secondly the co-unit, that is a natural transformation $\epsilon : U \cdot P \to 1$ such that for every object C in $\mathcal{B}$ and every f in $\mathsf{Hom}(U(C), D)$ we have $\epsilon_D \cdot U(\alpha_D(f)) = f$. Now we proceed by defining the syntactic ILT-indexed category starting from an ILT-theory.

Definition 12 *Given an ILT-theory $\mathcal{T}$ with any set of ground types $\mathcal{G}$ we define the* syntactic IL- indexed category *$F(\mathcal{T})$ in the following way.*

- Base category. *Objects of the base category $\mathcal{B}(\mathcal{T})$ are lists of types $(A_1, \ldots, A_n)$. The terminal object is the empty list $[\,]$.*

  Morphisms from $(A_1, \ldots, A_n)$ to $(B_1, \ldots, B_m)$ are lists of terms $(M_1, \ldots, M_m)$ such that

  $$x_1\colon A_1, \ldots, x_n\colon A_n \mid \_ \vdash M_i\colon B_i$$

  for some intuitionistic variables $x_1, \ldots, x_n$. We will write $\vec{A}$ for $(A_1, \ldots, A_n)$ whenever convenient.

  Two morphisms $(M_1, \ldots, M_m)$ and $(N_1, \ldots, N_m)$ from $(A_1, \ldots, A_n)$ to $(B_1, \ldots, B_m)$ supposing that $x_1\colon A_1, \ldots, x_n\colon A_n \mid \_ \vdash M_i\colon B_i$ and $y_1\colon A_1, \ldots, y_n\colon A_n \mid \_ \vdash N_i\colon B_i$ are equal if we derive

  $$x_1\colon A_1, \ldots, x_n\colon A_n | \_ \vdash M_i = N_i[\vec{x}/\vec{y}]\colon B_i$$

  We will write $(\vec{M})$ for $(M_1, \ldots, M_m)$ whenever convenient.

  The identity morphism on $(\vec{A})$ is the list $(\vec{x})$;

  Composition is given by intuitionistic substitution: given morphisms $(M_1, \ldots, M_m)$ from $(A_1, \ldots, A_n)$ to $(B_1, \ldots, B_m)$ with $x_1\colon A_1, \ldots, x_n\colon A_n \mid \_ \vdash M_i\colon B_i$ and $(N_1, \ldots, N_n)$ from $(C_1, \ldots, C_k)$ to $\vec{A}$ such that $y_1\colon C_1, \ldots, y_k\colon C_k \mid \_ \vdash N_j\colon A_j$, we define $\vec{M} \cdot \vec{N}$ to be $(M_1[\vec{N}/\vec{x}], \ldots, M_m[\vec{N}/\vec{x}])$.

- Fibres. *The objects of the fibres of* $E(\vec{A})$ *are types* A.

  A morphism from A *to* B *in* $E(\vec{A})$ *is a term* M *such that* $x_1\colon A_1,\ldots,x_n\colon A_n \mid a\colon A\vdash M\colon B$. *Two morphisms* M *and* N *from* A *to* B *in* $E(\vec{A})$ *such that* $x_1\colon A_1,\ldots,x_n\colon A_n \mid a\colon A\vdash M\colon B$ *and* $y_1\colon A_1,\ldots,y_n\colon A_n \mid b\colon A\vdash N\colon B$ *are equal if we derive* $\vec{x}\colon\vec{A}\mid a\colon A\vdash M = N[\vec{x}/\vec{y}, a/b]\colon B$.

  For any morphism $\vec{M}$ *from* $\vec{A}$ *to* $\vec{B}$, *the functor* $E(\vec{M})$ *is the identity on the objects and transforms any morphism* M *with* $\vec{y}\colon\vec{B}\mid a\colon A\vdash M\colon B$ *to* $M[\vec{M}/\vec{y}]$.

- Fibre structure. *The tensor product of two objects* A *and* B *in the fibre* $E(\vec{A})$ *is the type* $A\otimes B$. *The tensor product of two morphisms* M *and* N *in* $E(\vec{A})$ *is the term*

  $$\mathsf{let}\ z\ \mathsf{be}\ a\otimes b\ \mathsf{in}\ M\otimes N$$

  if $x_1\colon A_1,\ldots,x_n\colon A_n \mid a\colon A\vdash M\colon B$ *and* $y_1\colon A_1,\ldots,y_n\colon A_n \mid b\colon A\vdash N\colon B$.

  The unit of the tensor product in the category $E(\vec{A})$ *is given by the type* I.

  The right adjoint to the tensor product in $E(\vec{A})$ *is given by the natural transformation mapping the morphism* M *from* $C\otimes A$ *to* B *to* $\lambda a^A.M[c\otimes a/b]$ *where* $\vec{x}\colon\vec{A}\mid b\colon C\otimes A\vdash M\colon B$; *the co-unit is the natural transformation whose component at the object* B *is given by the morphism* $\mathsf{let}\ c\ \mathsf{be}\ a\otimes b\ \mathsf{in}\ ab$ *with* $\vec{x}\colon\vec{A}\mid c\colon (A\multimap B)\otimes A\vdash \mathsf{let}\ c\ \mathsf{be}\ a\otimes b\ \mathsf{in}\ ab\colon B$.

- Comprehension. *The right adjoint* P *to the functor* U *is given by*

  $$\begin{array}{rcl} P(((A_1,\ldots A_n),A)) & = & (A_1,\ldots,A_n,A) \\ P(((M_1,\ldots M_n),M)) & = & (M_1,\ldots,M_n,M[x/a]) \end{array}$$

  if $\vec{x}\colon\vec{A}\mid a\colon A\vdash M\colon B$, *since by lemma 1 we can derive* $\vec{x}\colon\vec{A},x\colon A\mid\ \vdash M[x/a]\colon B$. *For any morphism* $((M_1,\ldots M_n),M)$ *with* $\vec{x}\colon\vec{A}\mid a\colon I\vdash M\colon B$ *the natural isomorphism* $[\vec{M},M]$ *is* $(M_1,\ldots,M_n,M[\bullet/a])$. *The co-unit for the object* $((A_1,\ldots,A_n),A)$ *is the morphism* $((x_1,\ldots,x_n),\mathsf{let}\ a\ \mathsf{be}\ *\ \mathsf{in}\ x)$.

- Right adjoint to weakening. *The right adjoint to* $\mathsf{Fst}^*\colon E(\vec{A})\to E((\vec{A},A))$ *is the functor* $\Pi_A(-)\colon E((\vec{A},A))\to E(\vec{A})$ *which maps every object* C *of* $E((\vec{A},A))$ *to* $A\to C$ *and every morphism* $x\colon B\vdash M\colon C$ *in* $E((\vec{A},A))$ *to* $y\colon A\to B\vdash\lambda z^A.M[yz/x]$. *The natural transformation* Cur_I *maps the morphism* M *from* C *to* B *in* $E(\vec{A},A)$ *to* $\lambda x^A.M$ *if* $\vec{x}\colon\vec{A},x\colon A\mid c\colon C\vdash M\colon B$; *the co-unit is the natural transformation whose component at the object* B *is given by the term* ax *where* $\vec{x}\colon\vec{A},x\colon A\mid a\colon A\to B\vdash ax\colon B$.

Note the subtle difference in the definition of the base category and the fibre: we define objects in the base category to be lists of types, whereas objects in the fibre are singleton types. If we had products in the calculus as in [MdPR00], then we could have chosen an uniform definition and defined the objects of $\mathcal{B}$ to be types rather than lists of types.

In contrast we have no choice for the definition of the fibres but to use types as objects. The reason is that with the other choice there is no way of turning the fibre into a symmetric monoidal closed category, as there is no way of defining $C{-\!\circ}A\otimes B$ in terms of $C{-\!\circ}A$ and $C{-\!\circ}B$. This is not a problem for a cartesian closed category, as in this case we have $C \to (A \times B) \equiv (C \to A) \times (C \to B)$.

The key part of the completeness theorem is the following proposition, whose proof is a routine verification:

Proposition 13 *For any ILT-theory $\mathcal{T}$ the syntactic ILT-indexed category $F(\mathcal{T})$ is an ILT-indexed category.*

Proof. We have to verify that the structure given in Definition 12 is indeed an ILT-indexed category. As composition in $\mathcal{B}$ is given by intuitionistic substitution, it is easy to see that $\mathcal{B}$ is indeed a category and it has got a terminal object. Now we show that $E(\vec{A})$ is a symmetric monoidal closed category for each object $\vec{A}$ of $\mathcal{B}$. Again, as composition in $E(\vec{A})$ is given by linear substitution, it is easy to see that $E(\vec{A})$ is indeed a category. Next, we show that the operator $\otimes$ is indeed a tensor product. So assume we are given morphisms M_1, M_2, N_1 and N_2. We have to show that $(M_1 \otimes M_2) \cdot (N_1 \otimes N_2) = (M_1 \cdot N_1) \otimes (M_2 \cdot N_2)$. Assume that

$$x_1\colon A_1, \ldots, x_n\colon A_n \mid e\colon A\otimes B \vdash \mathsf{let}\ e\ \mathsf{be}\ a\otimes b\ \mathsf{in}\ M_1\otimes M_2\colon C\otimes D$$

and

$$x_1\colon A_1, \ldots, x_n\colon A_n \mid f\colon C\otimes D \vdash \mathsf{let}\ f\ \mathsf{be}\ c\otimes d\ \mathsf{in}\ N_1\otimes N_2\colon E\otimes F^3$$

Then we have

$$\begin{array}{rl} (\mathsf{let}\ e\ \mathsf{be}\ a\otimes b\ \mathsf{in}\ M_1\otimes M_2)[\mathsf{let}\ f\ \mathsf{be}\ c\otimes d\ \mathsf{in}\ N_1\otimes N_2/e] & = \\ \mathsf{let}\ (\mathsf{let}\ c\otimes d\ \mathsf{be}\ f\ \mathsf{in}\ N_1\otimes N_2)\ \mathsf{be}\ a\otimes b\ \mathsf{in}\ M_1\otimes M_2 & \stackrel{cc}{=} \\ \mathsf{let}\ f\ \mathsf{be}\ c\otimes d\ \mathsf{in}\ \mathsf{let}\ N_1\otimes N_2\ \mathsf{be}\ a\otimes b\ \mathsf{in}\ M_1\otimes M_2 & \stackrel{\beta}{=} \\ \mathsf{let}\ f\ \mathsf{be}\ c\otimes d\ \mathsf{in}\ M_1[N_1/a]\otimes M_2[N_2/b] & \end{array}$$

Next we verify that $\mathsf{Id}\otimes\mathsf{Id} = \mathsf{Id}$. This follows from an application of the η-rule for the tensor product:

$$\mathsf{let}\ c\ \mathsf{be}\ a\otimes b\ \mathsf{in}\ a\otimes b \quad \stackrel{\eta}{=} \quad c$$

[3] We have simplified the presentation here by using the same intuitionistic variables for showing the well-formedness of M_1, M_2, N_1 and N_2. To be precise one needs an extra substitution $N_1[\vec{x}/\vec{y}]$, $N_2[\vec{x}/\vec{y}]$, which we have omitted. We use this simplification throughout the proof.

Now we verify the adjunction defining linear function spaces. For this purpose we recall that in general to prove that the functors U and G form an adjunction $U \dashv G$, it suffices to check the following equations: for $t \in \mathsf{Hom}(U(A), D)$, $s : A' \to A$

$$\begin{array}{rcl} \epsilon_D \cdot U(\phi_{A,D}(t)) & = & t \\ \phi_{A,D}(t) \cdot s & = & \phi_{A',D}(t \cdot U(s)) \\ \phi_{G(D),D}(\epsilon_D) & = & \mathsf{Id}_{G(D)} \end{array}$$

where $\phi_{A,D} : \mathsf{Hom}(U(A), D) \to \mathsf{Hom}(A, G(D))$ is the isomorphism of the adjunction natural on A and D and $\epsilon : U \cdot G \to 1$ the corresponding counit. Now we check these equations in the case of the adjunction $(-) \otimes A \dashv A \multimap (-)$. The first equation is verified as follows:

$$\begin{array}{rcl} \mathsf{let}\ (\mathsf{let}\ b\ \mathsf{be}\ c \otimes d\ \mathsf{in}\ (\lambda a^A.M[c \otimes a/b]) \otimes d)\ \mathsf{be}\ a' \otimes b'\ \mathsf{in}\ a'b' & \stackrel{cc}{=} & \\ \mathsf{let}\ b\ \mathsf{be}\ c \otimes d\ \mathsf{in}\ \mathsf{let}\ (\lambda a^A.M[c \otimes a/b]) \otimes d\ \mathsf{be}\ a' \otimes b'\ \mathsf{in}\ a'b' & \stackrel{\beta}{=} & \\ \mathsf{let}\ b\ \mathsf{be}\ c \otimes d\ \mathsf{in}\ M[c \otimes d/b] & \stackrel{cc}{=} & \\ M[\mathsf{let}\ b\ \mathsf{be}\ c \otimes d\ \mathsf{in}\ c \otimes d/b] & \stackrel{\eta}{=} & M \end{array}$$

The second equation is verified as follows:

$$\begin{array}{rcl} (\lambda a^A.M[c \otimes a/b])[N/c] & = & \lambda a^A.M[N \otimes a/b] \\ & \stackrel{\beta}{=} & \lambda a^A.M[\mathsf{let}\ c' \otimes a\ \mathsf{be}\ c' \otimes a\ \mathsf{in}\ (N \otimes a)/b] \\ & = & \lambda a^A.M[\mathsf{let}\ d\ \mathsf{be}\ c' \otimes a\ \mathsf{in}\ (N \otimes a)/b][c' \otimes a/d] \end{array}$$

The third equation is verified as follows:

$$\lambda a^A.\mathsf{let}\ d\ \mathsf{be}\ b \otimes a'\ \mathsf{in}\ ba'[c \otimes a/d] \stackrel{\beta}{=} \lambda a^A.ca \stackrel{\eta}{=} c$$

The verifications for the unit type is similar and omitted. It remains to verify the equalities with respect to the right adjunction to the functor U. Here we only check the first of the three required equations, since the other ones follow easily by rules on substitution and β-equality for the let-expression regarding the type I.

$$\begin{array}{rcl} (\vec{x}, \mathsf{let}\ a\ \mathsf{be}\ *\ \mathsf{in}\ x)[\vec{M}/\vec{x}, M[*/a]/x] & = & (\vec{M}, \mathsf{let}\ a\ \mathsf{be}\ *\ \mathsf{in}\ M[*/a]) \\ & \stackrel{cc}{=} & (\vec{M}, M[\mathsf{let}\ a\ \mathsf{be}\ *\ \mathsf{in}\ */a]) \\ & \stackrel{\eta}{=} & (\vec{M}, M) \end{array}$$

□

Now we turn to the completeness theorem.

Theorem 14 *Consider the interpretation* $[\![\_]\!]$ *given in definition 9. If* $[\![\Gamma \mid \Delta \vdash M\colon A]\!] = [\![\Gamma \mid \Delta \vdash N\colon A]\!]$ *and for every IL- indexed category* $E\colon \mathcal{B}^{op} \to$ **Cat** *and for all derived sequents* $\Gamma \mid \Delta \vdash M\colon A$ *and* $\Gamma \mid \Delta \vdash N\colon A$*, then we can derive in ILT* $\Gamma \mid \Delta \vdash M = N\colon A$*.*

Proof. The proof immediately follow since the interpretation of ILT in the syntactic model out of ILT done in proposition 13 turns out to be the identity on the various judgements. □

Internal language. Starting from the above soundness and completeness theorems we want to see if ILT is actually an internal language of ILT-indexed categories. This means that the categories (*i.e.*,the models) and the type theory can be used interchangeably to reason about the type theory. This is a stronger property than soundness and completeness, which ensures only that the categorical structure is necessary and sufficient to interpret the type-theoretic constructors. ILT-theories will be an internal language only if we add to the type theories judgements of the form $A = B$, stating that two types A and B are equal. We also have the additional judgement

$$\frac{\Gamma \vdash M\colon A \qquad A = B}{\Gamma \vdash M\colon B}$$

To make the isomorphism between the models and the type theory precise we define the categories TH(ILT) and **ILT-ind**, which are the categories of type theories and categorical models respectively.

Definition 15 *The objects of* TH*(ILT) are the ILT-theories, i.e. type theories whose inference rules include the ILT ones. The morphisms are translations that send types to types so as to preserve*

$$I, 1, \otimes, \multimap, \to$$

They send terms to terms so as to preserve the introduction and elimination constructors corresponding to the above types and they send intuitionistic (linear) variables to intuitionistic (linear) variables such that the typability judgement and the equality between terms are preserved.

The definition of ILT-indexed categories will be given in two steps. First we define categorical structure which handles the two contexts and the relationship between them, and secondly we add structure which models the specific connectives of ILT.

The categorical structure for double contexts is given by the category of simply linear indexed categories:

Definition 16 *The objects of the category* **SL-ind** *are simply linear indexed categories and the morphisms between* $E\colon \mathcal{B}^{op} \to$ **Cat** *and* $E'\colon \mathcal{B}'^{op} \to$

Cat *are given by a functor* $H : \mathcal{B} \to \mathcal{B}'$ *preserving the terminal object and a natural transformation* $\alpha : E \Rightarrow E' \cdot H$ *such that for every object* Δ *in* $\mathcal{B}$ $\alpha_\Delta : E(\Delta) \to E'(H(\Delta))$ *is a SMC-functor. Finally, the comprehension adjunction is preserved and the intuitionistic function space too, as expressed by the conditions described in the following (where we differentiate the structure of* E *from that one of* E' *with the prime).*

1. *For every object* Δ *in* $\mathcal{B}$ *and* A *in* $E(\Delta)$*, and for every morphism* $(f,t) : (\Delta, A) \to (\Gamma, C)$ *in* $Gr(E)$ *we have* $H(P(\Delta, A)) = P'(H(\Delta), \alpha_\Delta(A))$ *and* $H(P(f,t)) = P'(H(f), \alpha_\Delta(t))$*.*

2. *For every morphism* $(f,t) : (\Delta, I) \to (\Gamma, C)$ *we have* $H([f,t]) = [(H(f), \alpha_\Delta(t))]'$

Then, based on this category we define the category of ILT-indexed categories as a subcategory:

Definition 17 *The category* **ILT-ind** *is the subcategory of* **SL-ind** *having as objects ILT-indexed categories and as morphisms between* $E \colon \mathcal{B}^{op} \to \mathbf{Cat}$ *and* $E' \colon \mathcal{B}'^{op} \to \mathbf{Cat}$ *morphisms* (H, α) *in* **SL-ind** *preserving the the intuitionistic function space, i.e., for every object* Δ *in* $\mathcal{B}$*,* A *in* $E(\Delta)$ *and* C *in* $E(\Delta.A)$ *and every morphism* f *in* $E(\Delta.A)$ *we have that* $\alpha_\Delta(\Pi_A(C)) = \Pi_{\alpha_\Delta(A)}(\alpha_{\Delta.A}(C))$ *and* $\alpha_\Delta(\mathsf{Cur}_I(f)) = \mathsf{Cur}_I{}'(\alpha_{\Delta.A}(f))$*.*

Note that in the definition of a morphism in **ILT-ind** the functor H on the base is actually *determined* by the natural transformation α because of the contextual base condition.

Formally the fact that ILT is the internal language of our ILT-indexed categories is proved by providing an equivalence between the category of ILT-theories TH(ILT) and the category of ILT-indexed categories **ILT-ind**. This equivalence describes how to use the categories (*i.e.*,the models) and the type theory interchangeably to reason about the type theory.
Now we are ready to prove the following:

Proposition 18 *There exist two functors* $L : \mathsf{TH}(ILT) \to$ **ILT-ind** *and* $F :$ **ILT-ind** $\to \mathsf{TH}(ILT)$ *that give rise to an equivalence between* $\mathsf{TH}(ILT)$ *and* **ILT-ind***.*

Proof. Given an ILT-theory $\mathcal{T}$ we define $F(\mathcal{T})$ in an analogous way to the definition of the syntactic ILT-category in Definition 12. In particular the context morphisms between the types A and C are $x : A \mid \_ \vdash c : C$. Hence we define $P((\_, B)) = B$ and $P((A, B)) = (A, B)$. We can easily see that $F(\mathcal{T})$ is an ILT-indexed category.
We can obviously lift any translation to become a morphism between ILT-indexed categories.
Given an ILT-indexed category $E \colon \mathcal{B}^{op} \to \mathbf{Cat}$ we now define an ILT-theory $L(E)$ out of it in the following way:

Definition 19 *Let $E\colon \mathcal{B}^{op} \to \mathbf{Cat}$ be an ILT-indexed category. The language of $L(E)$ is defined as follows:*

1. *the types of $L(E)$ are the objects of the fibre $E(\top_{\mathcal{B}})$;*
2. *the pre-terms of $L(E)$ are the morphisms of $E(\Delta)$ for every object Δ of $\mathcal{B}$;*
3. *The judgement $\Gamma|\Delta \vdash M\colon A$ holds in $L(E)$ if M is a morphism from Δ to A in $E(\Gamma)$;*
4. *Two types are equal if they are the same object in $L(E)$;*
5. *The judgement $\Gamma|\Delta \vdash M = N\colon A$ holds in $L(E)$ if M and N are equal morphisms in $E(\Gamma)$.*

The functor L can be easily extended on the morphisms of ILT-indexed categories to define translations.

This definition shows why the extension of ILT-theories with non-trivial equalities of types is be needed: in an arbitrary category, the object $A\otimes B$ might be equal to some other object C.

What remains to be checked is that the two compositions $L \cdot F$ and $F \cdot L$ are naturally isomorphic to the corresponding identity functors. For every ILT-theory $\mathcal{T}$ it is easy to check that $L(F(\mathcal{T}))$ can be translated into $\mathcal{T}$ via a natural isomorphism.

Vice versa for any ILT-indexed category $E\colon \mathcal{B}^{op} \to \mathbf{Cat}$ we prove that $F(L(E))$ is equivalent to E. The base category $\mathcal{B}$ is equivalent to $\mathcal{B}(F(L(E)))$ since by the comprehension adjunction with respect to E and the contextual base clause we can build a faithful, full and surjective functor from $\mathcal{B}(F(L(E)))$ towards $\mathcal{B}$ sending the empty context to $\top_{\mathcal{B}}$. The natural transformation on each fibre is given by suitable projecting functors on the objects and by the identity on morphisms. The components of this natural transformations are isomorphisms by the non-dependency clause. $\square$

Analogously, we can define the subcategory $\mathbf{ILT}_p$**-ind** of **ILT-ind** having as objects the indexed categories in **ILT-ind** whose fibres are SMCC with finite products and whose projecting functors preserve them and having as morphisms those of **ILT-ind** whose natural transformation are SMCP functors (i.e. functors preserving the SMC structure and finite products). Then, we can prove that ILT_p provides the internal language for the indexed categories in $\mathbf{ILT}_p$**-ind**, i.e. that $\mathsf{TH}(\mathrm{ILT})$ is equivalent to $\mathbf{ILT}_p$**-ind**.

Note that the internal language could be naturally enriched with explicit substitutions on terms to represent the composition in the fibre by explicit substitutions of linear variables and the morphism assignment of E by explicit substitutions of intuitionistic variables. But if we want to interpret explicit substitution operations on contexts in a different way from those on terms, then we need to add another fibration to each SMC fibre in the style of [GdPR99], passing to a complicated doubly indexed category.

9 The fibrational semantics for the other type theories

In this section we present the fibrational semantics for LNL and DILL. These semantics are based on the fibrational semantics for ILT.

9.1 The semantics for LNL

The fibrational semantics for LNLis given by an ILT-indexed category plus an adjunction which is the semantic counterpart of the bijection between set of terms M such that $\Gamma, x\colon \sigma|\Delta \vdash_{\mathcal{L}} M\colon A$ and the set of terms M' such that $\Gamma|\Delta, a\colon F(\sigma) \vdash_{\mathcal{L}} M'\colon A$ (Proposition 3). After showing that this semantics indeed captures LNLwe present an equivalent definition which requires a left adjoint to the inclusion functor modelling the inclusion of linear terms into intuitionistic terms.

9.1.1 The semantics via fibred adjunction

The precise definition of an LNL-indexed category is as follows:

Definition 20 *Let $E : \mathcal{B}^{op} \to \mathbf{Cat}$ be a simple linear indexed category. E is called an* LNL-category *if*

(i) *For every object Γ and X of $\mathcal{B}$ there exists an object $F(X)$, and the functor $\pi_X^*\colon E(\Gamma) \to E(\Gamma \times X)$ has a left adjoint F_X such that there exists an object $F(X)$ satisfying $F_X(A) = F(X) \otimes A$. If $f\colon A \to B$ is a morphism in $E(\Gamma \times X)$, we write $\nu_X(f)$ for the morphism in $E(\Gamma)$ obtained by transposing f across the adjunction.*

(ii) *Moreover, the Beck-Chevalley condition for the adjunction $F_X \vdash \pi_X^*$ is satisfied in the strict sense, i.e. the equations*

$$f^* \cdot F_X = F_X \cdot (f.\mathsf{Id})^* \qquad f^*(\nu_X(g)) = \nu_X((f.\mathsf{Id})^*(g))$$

hold for every $f\colon \Delta \to \Gamma$, $A \in ObE(\Gamma)$, $B \in ObE(\Gamma)$ and $g \in \mathrm{Hom}_{E(\Gamma\times X)}(A, B)$.

(iii) *For each morphism $f\colon A \to B$ in $E(\Gamma \times X)$, we have*

$$\nu_X(f \otimes \mathsf{Id}) = \nu_X(f) \otimes \mathsf{Id}$$

and we have also

$$F_X(r) = r$$

where r is the morphism $A \otimes I \to A$.

We will omit the subscript in F_X, η_X and π_X if it is clear from the context.

In LNL, the intuitionistic implication $A \to B$ can be translated as a linear type as $FGA \multimap B$ and this has a semantic counterpart exactly as follows:

Proposition 21 *Consider an* **LNL-ind**-*category* $E\colon \mathcal{B}^{op} \to \mathbf{Cat}$. *For every object* Γ *of* $\mathcal{B}$ *and* A *of* $E(\Gamma)$, *the functor* $\mathsf{Fst}^*_A\colon E(\Gamma) \to E(\Gamma.A)$ *has a right adjoint* $\Pi_A\colon E(\Gamma.A) \to E(\Gamma)$. *We will write in the sequel* Cur_I *for the natural isomorphism between* $\mathrm{Hom}_{E(\Gamma.A)}(\mathsf{Fst}^*_A(B), C)$ *and* $\mathrm{Hom}_{E(\Gamma)}(B, \Pi_A(C))$ *and* App_I *for its counit. Moreover, the Beck-Chevalley condition for the adjunction* $\mathsf{Fst}^*_A \vdash \Pi_A$ *is satisfied in the strict sense, i.e. the equations*

$$f^* \cdot \Pi_A = \Pi_A \cdot (f.\mathsf{Id})^* \qquad f^*(\mathsf{Cur}_I(g)) = \mathsf{Cur}_I((f.\mathsf{Id})^*(g))$$

hold for every $f\colon \Delta \to \Gamma$, $A, B, C \in ObE(\Gamma)$, *and* $g \in \mathrm{Hom}_{E(\Gamma.A)}(B, C)$.

Proof. Define the right adjoint to Fst^*_A by $\Pi_A(B) = F(\top.A) \multimap B$. The desired isomorphism can be seen by

$$\begin{array}{rcl} \mathsf{Hom}_{E(\Gamma.A)}(C, B) & \cong & \mathsf{Hom}_{E(\Gamma \times (\top.A))}(C, B) \\ & \cong & \mathsf{Hom}_{E(\Gamma)}(F(\top.A) \otimes C), B)\ ) \\ & \cong & \mathsf{Hom}_{E(\Gamma)}(C, F(\top.A) \multimap B) \end{array}$$

The naturality in the second component follows directly from the naturality of the bijections involved. The Beck-Chevalley-condition is a direct consequence of the Beck-Chevalley-condition for the functor F.

Explicitly, we have

$$\mathsf{Cur}_I(f) = \mathsf{Cur}_L(\nu(f))$$

$\square$

Now we show to interpret LNL in any **LNL-ind**-category.

Definition 22 *Given any* **LNL-ind**-*category* $E\colon \mathcal{B}^{op} \to \mathbf{Cat}$, *we define a map* $[\![\_]\!]$ *from types to objects in* $E(\top_{\mathcal{B}})$, *from intuitionistic contexts* Γ *to objects in* $\mathcal{B}$, *from linear contexts* Δ *to objects in* $E(\top_{\mathcal{B}})$, *from double contexts to objects of suitable fibres, from terms* $\Gamma \mid \Delta \vdash_{\mathcal{L}} M\colon A$ *to morphisms* $[\![M]\!]\colon [\![\Delta]\!] \to [\![A]\!]$ *in* $E([\![\Gamma]\!])$ *and from terms* $\Gamma \vdash_{\mathcal{C}} t\colon \sigma$ *to morphisms* $[\![t]\!]\colon [\![\Gamma]\!] \to [\![\sigma]\!]$ *in* $\mathcal{B}$ *as follows (assuming that all* f^* *are identities on the objects):*

- *(i) For all constructors of LNL except* F *and* G, *the interpretation of the corresponding constructors in LNL is given by the clauses for ILT;*
- *(ii) On types:* $[\![F(\sigma)]\!] = F([\![\sigma]\!]) \qquad [\![G(A)]\!] = \top_{\mathcal{B}}.[\![A]\!]$

(iii) On terms:

$$\frac{[\![\Gamma \vdash_{\mathcal{C}} t : \sigma]\!] = h}{[\![\Gamma \mid \_ \vdash_{\mathcal{L}} F(t) : F(\sigma)]\!] = r \cdot [\mathsf{Id}, h]^* \eta_I}$$

$$\frac{[\![\Gamma|\Delta_1 \vdash_{\mathcal{L}} M \colon F(\sigma)]\!] = f \qquad [\![\Gamma, x \colon \sigma|\Delta_2 \vdash_{\mathcal{L}} N \colon B]\!] = g}{[\![\Gamma|\Delta \vdash_{\mathcal{L}} \mathsf{let}\ M\ \mathsf{be}\ F(x)\ \mathsf{in}\ N \colon B]\!] = \nu_{[\![\sigma]\!]}(g) \cdot (f \otimes \mathsf{Id}_{\Delta_2}) \cdot c}$$

$$\frac{\Gamma|\_ \vdash_{\mathcal{L}} M \colon A}{[\![\Gamma \vdash_{\mathcal{C}} G(M) \colon G(A)]\!] = [\langle\rangle, [\![M]\!]]}$$

$$\frac{[\![\Gamma \vdash_{\mathcal{C}} t \colon G(A)]\!] = h}{[\![\Gamma|\_ \vdash_{\mathcal{L}} \mathsf{derelict}(t) \colon A]\!] = h^*(\mathsf{Snd})}$$

where c is the canonical morphism from $[\![\Delta]\!]$ to $[\![\Delta_1]\!] \otimes [\![\Delta_2]\!]$, η and ν are the unit and one part of the bijection between Hom-sets of the adjunction in Definition 20.

Note that the term constructors F and G are not interpreted by an adjunction, as in LNL. The reason is that this adjunction is split into two adjunctions in our semantics: the comprehension restricted to $E(\top)$ (which describes how intuitionistic and linear terms interact and models the type constructor G and the term constructor $\mathsf{derelict}(\_)$) and the adjunction of definition 20 which provides linear type constructors for intuitionistic contexts.

Now we turn to soundness. The substitution lemmata for ILT carry over to LNL.

Lemma 23 *(i) Assume $[\![\Gamma, x \colon \sigma \mid \Delta \vdash_{\mathcal{L}} M \colon B]\!] = f$ and $[\![\Gamma \vdash_{\mathcal{C}} t \colon \sigma]\!] = g$. Then $[\![\Gamma \mid \Delta \vdash M[t/x] \colon B]\!] = [\mathsf{Id}, g]^*(f)$.*

(ii) Assume $[\![\Gamma \mid \Delta_1, a \colon A \vdash M \colon B]\!] = f$ and $[\![\Gamma \mid \Delta_2 \vdash N \colon A]\!] = g$. Then $[\![\Gamma \mid \Delta \vdash M[N/a] \colon B]\!] = f \cdot (\mathsf{Id} \otimes g) \cdot c$, where c is the canonical morphism from $[\![\Delta]\!]$ to $[\![\Delta_1 \otimes \Delta_2]\!]$.

Proof. We use an induction over the structure of M and t. The only interesting cases are the ones involving the term constructors F and G. The cases of substitution for intuitionistic variables in $F(t)$ and $\mathsf{let}\ M\ \mathsf{be}\ F(y)\ \mathsf{in}\ N$ use the Beck-Chevalley-condition of the adjunction defining the functor F. □

Theorem 24 *Given an* **LNL-ind***-category $E \colon \mathcal{B}^{op} \to$* **Cat** *under the above interpretation $[\![]\!]$ the following facts hold.*

(i) Assume $\Gamma \mid \Delta \vdash_{\mathcal{L}} M \colon A$. Then $[\![\Gamma \mid \Delta \vdash_{\mathcal{L}} M \colon A]\!]$ is a morphism from $[\![\Delta]\!]$ to $[\![A]\!]$ in $E([\![\Gamma]\!])$;

(ii) Assume $\Gamma \vdash_{\mathcal{C}} t \colon \sigma$. Then $[\![\Gamma \vdash_{\mathcal{C}} t \colon \sigma]\!]$ is a morphism from $[\![\Gamma]\!]$ to $[\![\sigma]\!]$ in $\mathcal{B}$;

(iii) Assume $\Gamma \mid \Delta \vdash_{\mathcal{L}} M = N\colon A$*. Then* $[\![\Gamma \mid \Delta \vdash_{\mathcal{L}} M\colon A]\!] = [\![\Gamma \mid \Delta \vdash_{\mathcal{L}} N\colon A]\!]$*;*

(iv) Assume $\Gamma \vdash_{\mathcal{C}} t = s\colon \sigma$*. Then* $[\![\Gamma \vdash_{\mathcal{C}} t\colon \sigma]\!] = [\![\Gamma \vdash_{\mathcal{C}} s\colon \sigma]\!]$*.*

Proof. We present here only the most complicated cases, which are the $F\beta$- and $F\eta$-equality. Let the morphism $h\colon [\![\Gamma]\!] \to [\![\sigma]\!]$ in $\mathcal{B}$ be the semantics of t. Let furthermore by $g\colon [\![\Delta]\!] \to [\![B]\!]$ in $E([\![\Gamma]\!] \times [\![\sigma]\!])$ be the semantics of M. By part (i) the semantics of the term $\mathsf{let}\ F(t)\ \mathsf{be}\ F(x)\ \mathsf{in}\ M$ is the morphism

$$\begin{aligned}
f &:= \nu_{[\![\sigma]\!]}(g) \cdot ((r \cdot [\mathsf{Id}, h]^* \eta_I) \otimes \mathsf{Id}_{[\![\Delta_2]\!]}) \cdot l^{-1} \\
&= [\mathsf{Id}, h]^* (\mathsf{Fst}^* \nu_{[\![\sigma]\!]}(g) \cdot (r \otimes \mathsf{Id}_{[\![\Delta_2]\!]}) \cdot (\eta_I \otimes \mathsf{Id}_{[\![\Delta_2]\!]}) \cdot l^{-1}) \\
&= [\mathsf{Id}, h]^* (\mathsf{Fst}^* \nu_{[\![\sigma]\!]}(g) \cdot \eta_{[\![\Delta_2]\!]}) \\
&= [\mathsf{Id}, h]^*(g) = [\mathsf{Id}, h]^*([\![M]\!])
\end{aligned}$$

The next case is the $F\eta$-rule. Let the morphism $g : [\![\Delta_1]\!] \to F([\![\sigma]\!])$ in $E(\Gamma)$ be the semantics of N, and $f : [\![\Delta_2]\!] \otimes [\![A]\!] \to [\![B]\!]$ be the semantics of M. Then the semantics of $\mathsf{let}\ N\ \mathsf{be}\ F(x)\ \mathsf{in}\ M[F(x)/a]$ is the morphism

$$\begin{aligned}
h &:= \nu_{[\![\sigma]\!]}(\mathsf{Fst}^*(f) \cdot (r \cdot [\mathsf{Id}, \mathsf{Snd}]^*(\eta_I) \otimes \mathsf{Id}_{\Delta_2}) \cdot l^{-1})) \cdot (g \otimes \mathsf{Id}_{\Delta_2}) \\
&= f \cdot \nu_{[\![\sigma]\!]}((r \cdot \eta_I) \otimes \mathsf{Id}_{\Delta_2} \cdot l^{-1}))) \cdot (g \otimes \mathsf{Id}_{\Delta_2}) \\
&= f \cdot (g \otimes \mathsf{Id}_{\Delta_2)}
\end{aligned}$$

$\square$

LNL is also the internal language of LNL-indexed categories. The development for ILT carries through in exactly the same way. Firstly, we define the category of LNL-indexed categories:

Definition 25 *The category* **LNL-ind** *is the subcategory of* **SL-ind** *having as objects LNL-indexed categories and as morphisms between* $E\colon \mathcal{B}^{op} \to$ **Cat** *and* $E'\colon \mathcal{B}'^{op} \to$ **Cat** *morphisms* (H, α) *in* **SL-ind** *such that the conditions described in the following are satisfied:*

1. *H preserves strictly the cartesian closed structure.*
2. *For every object* Γ *and* X *of* $\mathcal{B}$*, the functors* $\alpha_{\Gamma \times X}$ *and* α_Γ *are the components of a transformation of adjoints.*

Now the fact that LNL is the internal language of LNL-indexed categories can be stated where the category $\mathsf{TH}(\mathrm{LNL})$ of LNL-theories is defined analogously to definition 15:

Proposition 26 *There exist two functors* $L : \mathsf{TH}(\mathit{LNL}) \to$ **LNL-ind** *and* $F :$ **LNL-ind** $\to \mathsf{TH}(\mathit{LNL})$ *that give rise to an equivalence between* $\mathsf{TH}(\mathit{LNL})$ *and* **LNL-ind***.*

Analogously, we can define **LNL$_p$-ind** as the subcategory of **LNL-ind** having as objects those indexed categories whose fibres are SMCC with finite products and whose projecting functors preserve them and having as morphisms those of **LNL-ind** whose natural transformations are componentwise SMCP functors. Then, we can prove that $_\&$LNL provides the internal languages for the indexed categories in **LNL$_p$-ind**.

9.1.2 A left adoint to the inclusion of linear terms into intuitionistic terms

We finish this section by giving an alternative definition of an LNL-category, which requires a left adjoint to the inclusion functor between the base category modelling intuitionistic terms and the category $E(\top)$ which models all the linear terms. The coherence conditions and Beck-chevalley condition of definition 20 turn out to be equivalent to the requirement that this adjunction is monoidal.

Proposition 27 *Let $E\colon \mathcal{B}^{op} \to$ **Cat** be a simple indexed category. Then the following are equivalent:*

(i) E is an LNL-category;

(ii) The functor $\mathcal{I}_\mathcal{B}\colon E(\top) \to \mathcal{B}$ which maps an object A to $\top.A$ and a morphism f to $\mathsf{Id}.f$ has a left adjoint $F_\mathcal{B}$, and this adjunction is monoidal.

We will write $\eta^\mathcal{B}$ and $\epsilon^\mathcal{B}$ for the unit and co-unit of the adjunction $F_\mathcal{B} \dashv \mathcal{I}_B$.

Before we give the proof of this proposition we show a lemma about sufficient conditions for an adjunction to be monoidal.

Lemma 28 *Let $F \dashv G$ be an adjunction between two symmetric monoidal category and let G be a monoidal functor. If there exists a map $m_F\colon F(A)\otimes F(B) \to F(A\otimes B)$ and the equation*

$$\eta_{A\otimes B} = G(m_F) \cdot m_G \cdot (\eta_A \otimes \eta_B)$$

is satisfied, then the adjunction is monoidal.

Proof. We show that m_F is a natural transformation as follows:

$$\begin{array}{rrcl}
 & F(f\otimes g)\cdot m_F & = & m_F \cdot F(f)\otimes F(g) \\
\iff & F(f\otimes g) & = & m_F \cdot F(f)\otimes F(g) \cdot m_F^{-1} \\
\iff & GF((f\otimes g))\cdot \eta & = & Gm_F \cdot G(F(f)\otimes F(g)) \cdot Gm_F^{-1} \cdot \eta \\
\iff & \eta \cdot (f\otimes g) & = & Gm_F \cdot G(F(f)\otimes F(g)) \cdot Gm_F^{-1} \cdot Gm_F \cdot m_G \cdot (\eta\otimes\eta) \\
 & & = & Gm_F \cdot m_G \cdot G(F(f))\otimes G(F(g)) \cdot (\eta\otimes\eta) \\
 & & = & Gm_F \cdot m_G \cdot \eta\otimes\eta) \cdot (f\otimes g) \\
 & & = & \eta \cdot (f\otimes g)
\end{array}$$

□

Now we turn to the proof of Proposition 27.

Proof. (*i*) **implies** (*ii*)**:** We have the following bijection between Hom-sets:

$$\frac{\dfrac{\dfrac{\mathsf{Hom}_{E(\top)}(F(\Gamma), B)}{\mathsf{Hom}_{E(\top)}(F(\Gamma)\otimes I, B)}}{\mathsf{Hom}_{E(\Gamma)}(I, B)}}{\mathsf{Hom}_{\mathcal{B}}(\Gamma, \top.B)}$$

The naturality in the second component follows directly from the naturality of the adjunction between $E(\top)$ and $E(\Gamma)$.

To see that the adjunction is monoidal, we show first that

$$\eta^{\mathcal{B}}_{\Gamma\otimes\Delta} = (\mathsf{Id}.m_{F_{\mathcal{B}}}) \cdot [\langle\rangle, \pi^*\mathsf{Snd}\otimes\pi'^*\mathsf{Snd}] \cdot (\eta^{\mathcal{B}}_{\Gamma}\otimes\eta^{\mathcal{B}}_{\Delta}) \cdot r^{-1}$$

holds, where $m_{F_{\mathcal{B}}}$ is defined as the morphism $\nu_\Gamma(\nu_\Delta(\nu^{-1}_{\Gamma\times\Delta}(r)))$. By inserting the definition of $\eta^{\mathcal{B}}$, the RHS, called g, of the above equation satisfies

$$\begin{aligned}
g^*\mathsf{Snd} &= \langle\rangle^*(\nu_\Gamma(\nu_\Delta(\nu^{-1}_{\Gamma\times\Delta}(r)))) \cdot r^{-1} \cdot (\pi^*(\nu^{-1}_{\Gamma}(r))\otimes\pi'^*(\nu^{-1}_{\Delta}(r))) \cdot r^{-1} \\
&= \langle\rangle^*(\nu_\Gamma(\nu_\Delta(\nu^{-1}_{\Gamma\times\Delta}(r)))) \cdot r^{-1} \cdot \nu^{-1}_{\Gamma}(\mathsf{Id}\otimes r) \cdot \nu^{-1}_{\Delta}(r\otimes\mathsf{Id}) \cdot r^{-1} \\
&= \langle\rangle^*(\nu_\Gamma(\nu_\Delta(\nu^{-1}_{\Gamma\times\Delta}(r)))) \cdot \nu^{-1}_{\Gamma}(\mathsf{Id}) \cdot \nu^{-1}_{\Delta}(\mathsf{Id}) \\
&= \nu^{-1}_{\Gamma}(\nu_\Gamma(\langle\rangle^*\nu_\Delta(\nu^{-1}_{\Gamma\times\Delta}(r)))) \cdot \nu^{-1}_{\Delta}(\mathsf{Id}) \\
&= \langle\rangle^*\nu_\Delta(\nu^{-1}_{\Gamma\times\Delta}(r)) \cdot \nu^{-1}_{\Delta}(\mathsf{Id}) \\
&= \nu^{-1}_{\Delta}(\nu_\Delta(\langle\rangle^*\nu^{-1}_{\Gamma\times\Delta}(r))) \\
&= \nu^{-1}_{\Gamma\times\Delta}(r) = {\eta^{\mathcal{B}}_{\Gamma\times\Delta}}^*\mathsf{Snd}
\end{aligned}$$

which was to be shown. Next, we show that m_F is a natural transformation, *i.e.*,that the diagram

$$\begin{tikzcd}
F(\Gamma_1)\otimes F(\Gamma_2) \arrow[r, "r^{-1}"] \arrow[d, "r^{-1}\otimes r^{-1}"] & F(\Gamma_1)\otimes F(\Gamma_2)\otimes I \arrow[r, "\nu_{\Gamma_1}(\nu_{\Gamma_2}(\nu^{-1}_{\Gamma_1\times\Gamma_2}(r)))"] & F(\Gamma_1\times\Gamma_2) \arrow[d, "r^{-1}"] \\
F(\Gamma_1)\otimes I\otimes F(\Gamma_2)\otimes I \arrow[d, "\nu_{\Gamma_1}(f_1^*(\nu_{\Delta_1}(r)))\otimes\nu_{\Gamma_1}(f_2^*(\nu_{\Delta_2}(r)))"] & & F(\Gamma\times\Gamma_2)\otimes I \arrow[d, "\nu_{\Gamma_1\times\Gamma_2}((f_1\times f_2)^*(\nu^{-1}_{\Delta_1\times\Delta_2}(r)))"'] \\
F(\Delta_1)\otimes F(\Delta_2) \arrow[r, "r^{-1}"] & F(\Delta_1)\otimes F(\Delta_2) \arrow[r, "\nu_{\Delta_1}(\nu_{\Delta_2}(\nu^{-1}_{\Delta_1\times\Delta_2}(r)))"] & F(\Delta_1\times\Delta_2)
\end{tikzcd}$$

commutes. The East-North direction is equal to the morphism

$$
\begin{aligned}
\nu_{\Gamma_1\times\Gamma_2}((f_1\times f_2)^*(\nu^{-1}_{\Delta_1\times\Delta_2}(r)))\cdot r^{-1}\cdot\nu_{\Gamma_1}(\nu_{\Gamma_2}(\nu^{-1}_{\Gamma_1\times\Gamma_2}(r)))\cdot r^{-1} &= \\
\nu_{\Gamma_1\times\Gamma_2}((f_1\times f_2)^*(\nu^{-1}_{\Delta_1\times\Delta_2}(r)))\cdot\nu_{\Gamma_1}(\nu_{\Gamma_2}(r^{-1}\cdot\nu^{-1}_{\Gamma_1\times\Gamma_2}(r)))\cdot r^{-1} &= \\
\nu_{\Gamma_1\times\Gamma_2}((f_1\times f_2)^*(\nu^{-1}_{\Delta_1\times\Delta_2}(r)))\cdot\nu_{\Gamma_1}(\nu_{\Gamma_2}(\nu^{-1}_{\Gamma_1\times\Gamma_2}(\mathsf{Id})))\cdot r^{-1} &= \\
\nu_{\Gamma_1}(\nu_{\Gamma_2}((\mathsf{Fst}\cdot\mathsf{Fst})^*(\nu_{\Gamma_1\times\Gamma_2}((f_1\times f_2)^*(\nu^{-1}_{\Delta_1\times\Delta_2}(r)))\cdot\nu^{-1}_{\Gamma_1\times\Gamma_2}(\mathsf{Id}))))\cdot r^{-1} &= \\
\nu_{\Gamma_1}(\nu_{\Gamma_2}(\mathsf{Fst}^*(\nu_{\Gamma_1\times\Gamma_2}((f_1\times f_2)^*(\nu^{-1}_{\Delta_1\times\Delta_2}(r)))\cdot\nu^{-1}_{\Gamma_1\times\Gamma_2}(\mathsf{Id}))))\cdot r^{-1} &= \\
\nu_{\Gamma_1}(\nu_{\Gamma_2}(\nu^{-1}_{\Gamma_1\times\Gamma_2}(\nu_{\Gamma_1\times\Gamma_2}((f_1\times f_2)^*(\nu^{-1}_{\Delta_1\times\Delta_2}(r))))))\cdot r^{-1} &= \\
\nu_{\Gamma_1}(\nu_{\Gamma_2}((f_1\times f_2)^*(\nu^{-1}_{\Delta_1\times\Delta_2}(r))))\cdot r^{-1} &
\end{aligned}
$$

The South-East direction is equal to

$$
\begin{aligned}
\nu_{\Delta_1}(\nu_{\Delta_2}(\nu^{-1}_{\Delta_1\times\Delta_2}(r)))\cdot r^{-1}\cdot\nu_{\Gamma_1}((\mathsf{Id}\times f_1)^*(\nu_{\Delta_1}(r)))\otimes\nu_{\Gamma_1}((\mathsf{Id}\times f_2)^*(\nu_{\Delta_2}(r)))\cdot(r^{-1}\otimes r^{-1}) &= \\
\nu_{\Delta_1}(\nu_{\Delta_2}(\nu^{-1}_{\Delta_1\times\Delta_2}(r)))\cdot r^{-1}\cdot\nu_{\Gamma_1}((\mathsf{Id}\times f_1)^*(\nu_{\Delta_1}(r)))\otimes\nu_{\Gamma_1}((\mathsf{Id}\times f_2)^*(\nu_{\Delta_2}(r)))\cdot(r^{-1}\otimes r^{-1}) &= \\
\nu_{\Delta_1}(\nu_{\Delta_2}(\nu^{-1}_{\Delta_1\times\Delta_2}(r)))\cdot(\mathsf{Id}\otimes\nu_{\Gamma_2}((r^{-1}\cdot(\mathsf{Id}\times f_2)^*(\nu^{-1}_{\Delta_2}(r))))\cdot & \\
(\nu_{\Gamma_1}(r^{-1}\cdot(\mathsf{Id}\times f_1)^*(\nu^{-1}_{\Delta_1}(r)))\otimes\mathsf{Id})\cdot(r^{-1}\otimes\mathsf{Id})\mathsf{Id}) &= \\
\nu_{\Delta_1}(\nu_{\Delta_2}(\nu^{-1}_{\Delta_1\times\Delta_2}(r)))\cdot(\mathsf{Id}\otimes\nu_{\Gamma_2}((\mathsf{Id}\times f_2)^*(\nu^{-1}_{\Delta_2}(\mathsf{Id})))\cdot & \\
(\nu_{\Gamma_1}((\mathsf{Id}\times f_1)^*(\nu^{-1}_{\Delta_1}(\mathsf{Id})))\otimes\mathsf{Id})\cdot(r^{-1}\otimes\mathsf{Id}) &= \\
\nu_{\Delta_1}(\nu_{\Delta_2}(\nu^{-1}_{\Delta_1\times\Delta_2}(r)))\cdot(\mathsf{Id}\otimes\nu_{\Gamma_2}((\mathsf{Id}\times f_2)^*(\nu^{-1}_{\Delta_2}(\mathsf{Id})))\cdot & \\
(\nu_{\Gamma_1}((\mathsf{Id}\times f_1)^*(\nu^{-1}_{\Delta_1}(\mathsf{Id})))\otimes\mathsf{Id})\cdot(r^{-1}\otimes\mathsf{Id}) &=
\end{aligned}
$$

Next, we show the nullary version:

(ii) **implies** (i)**:** We show this result in two steps. Firstly, we show that the functor $\mathcal{I}\colon Gr(E)\to E(\top)$ has a monoidal left adoint (Lemma 29), and secondly we use this lemma to show the claim (Lemma 30).

□

Lemma 29 *Let $E\colon\mathcal{B}^{op}\to\mathbf{Cat}$ be a simple indexed category, and assume the functor $\mathcal{I}_{\mathcal{B}}\colon E(\top)\to\mathcal{B}$ which sends an object A to the object $\top.A$ and a morphism f to the morphism $[\mathsf{Id},f]$ has a left adjoint $F_{\mathcal{B}}$, and this adjunction is monoidal. Then the functor $\mathcal{I}\colon E(\top)\to Gr(E)$ which sends an object A to $(\top,A)$ and a morphism f to (Id,f) has a left adjoint $\hat{F}$, and this adjunction is monoidal.*

Proof. Define $\hat{F}((\Gamma,A)) = F(\Gamma)\otimes A$. We have the following bijections between Hom-sets:

$$\begin{array}{c}
\mathsf{Hom}_{Gr(E)}((\Gamma, A), (\top, B)) \\
\hline\hline
\mathsf{Hom}_{Gr(E)}((\Gamma, I), (\top, A \multimap B)) \\
\hline\hline
\mathsf{Hom}_{\mathcal{B}}(\Gamma, A \multimap B) \\
\hline\hline
\mathsf{Hom}_{E(\top)}(F(\Gamma), A \multimap B) \\
\hline\hline
\mathsf{Hom}_{E(\top}(F(\Gamma) \otimes A, B)
\end{array}$$

The naturality in the second component follows from the fact that the following diagram commutes:

$$\begin{tikzcd}
\mathsf{Hom}_{Gr(E)}((Y, A), (\top, B)) \arrow[r, "(\mathsf{Id}, g)_*"] \arrow[d] & \mathsf{Hom}_{Gr(E)}((Y, A), (\top, C)) \arrow[d] \\
\mathsf{Hom}_{Gr(E)}((Y, I), (T, A \multimap B)) \arrow[r, "(\mathsf{Id}, \mathsf{Cur}(g \cdot \mathsf{App}))_*"] \arrow[d] & \mathsf{Hom}_{Gr(E)}((Y, I), (T, A \multimap C)) \arrow[d] \\
\mathsf{Hom}_{\mathcal{B}}(Y, \top . A \multimap B) \arrow[r, "\langle \langle \rangle, \mathsf{Cur}(g \cdot \mathsf{App}) \rangle_*"] \arrow[d] & \mathsf{Hom}_{\mathcal{B}}(Y, \top . A \multimap C) \arrow[d] \\
\mathsf{Hom}_{E(\top)}(F(Y), A \multimap B) \arrow[r, "\mathsf{Cur}(g \cdot \mathsf{App})_*"] \arrow[d] & \mathsf{Hom}_{E(\top)}(F(Y), A \multimap C) \arrow[d] \\
\mathsf{Hom}_{E(\top)}(F(Y) \otimes A, B) \arrow[r, "g_*"] & \mathsf{Hom}_{E(\top)}(F(Y) \otimes A, C)
\end{tikzcd}$$

where the vertical arrows are the corresponding part of the bijection of the hom-sets and g_* is the usual function on hom-sets which sends a morphism f to the morphism $g \circ f$.

To see that this adjunction is monoidal, we show first by transposing the identity over the adjunction that $\hat{\eta}_{(\Gamma, A)}$ is the morphism

$$(\Gamma, A) \xrightarrow{(\mathsf{Id}, l^{-1})} (\Gamma, I \otimes A) \xrightarrow{(\langle \rangle, \eta_{\Gamma}^{\mathcal{B}^*} \mathsf{Snd} \otimes id)} (\top, F(\Gamma) \otimes A)$$

Now a routine calculation shows that the following diagram commutes:

$$\begin{tikzcd}
(\Gamma, A)\otimes(\Delta, B) \arrow[r, "\hat{\eta}_{(\Gamma,A)}\otimes\hat{\eta}_{(\Delta,B)}"] \arrow[d, "\hat{\eta}"'] & (\top, F(\Gamma, A))\otimes(\top, F(\Delta, B)) \arrow[d, "m_{\mathcal{I}}"] \\
(\top, F(\Gamma\otimes\Delta)\otimes A\otimes B) & (\top, F(\Gamma)\otimes F(\Delta)\otimes A\otimes B) \arrow[l, "(\mathsf{Id}, m_{F_{\mathcal{B}}}\otimes\mathsf{Id})"]
\end{tikzcd}$$

The fact that $\hat{F}$ is monoidal follows now follows from Lemma 28. $\square$

Lemma 30 *The following bijection between hom-sets*

$$\frac{\frac{\frac{\frac{\frac{\mathsf{Hom}_{E(\Gamma\times Y)}(A, B)}{\mathsf{Hom}_{Gr(E)}((\Gamma\times Y, A), (\top, B))}}{\mathsf{Hom}_{E(\top)}(F(\Gamma\times Y)\otimes A, B)}}{\mathsf{Hom}_{E(\top)}(F(\Gamma)\otimes F(Y)\otimes A, B)}}{\mathsf{Hom}_{Gr(E)}((\Gamma, F(Y)\otimes A), (\top, B)}}{\mathsf{Hom}_{E(\Gamma)}(F(Y)\otimes A, B)}$$

where the bijection between the second and third line is given by the adjunction of lemma 29, defines a left adjoint to the functor Fst_Y *which satisfies the Beck-Chevalley condition.*

Proof. It is easy to see that the diagram in lemma 30 defines indeed a bijection. To show naturality in the second component we show that the following diagram commutes:

$$\begin{tikzcd}
\mathsf{Hom}_{E(\Gamma\times Y)}(A,B) \arrow[r, "(\mathsf{Fst}^*g)_*"] \arrow[d] \arrow[dr, phantom, "(1)"] & \mathsf{Hom}_{E(\Gamma\times Y)}(A,C) \arrow[d] \\
\mathsf{Hom}_{Gr(E)}((\Gamma\times Y,A),(\top,B)) \arrow[r, "\kappa_1"] \arrow[d] \arrow[dr, phantom, "(2)"] & \mathsf{Hom}_{Gr(E)}((\Gamma\times Y,A),(\top,C)) \arrow[d] \\
\mathsf{Hom}_{E(\top)}(F(\Gamma\times Y)\otimes A,B) \arrow[r, "\kappa_2"] \arrow[d] \arrow[dr, phantom, "(3)"] & \mathsf{Hom}_{E(\top)}(F(\Gamma\times Y)\otimes A,C) \arrow[d] \\
\mathsf{Hom}_{E(\top)}(F(\Gamma)\otimes F(Y)\otimes A,B) \arrow[r, "\kappa_3"] \arrow[d] \arrow[dr, phantom, "(4)"] & \mathsf{Hom}_{E(\top)}(F(\Gamma)\otimes F(Y)\otimes A,C) \arrow[d] \\
\mathsf{Hom}_{Gr(E)}((\Gamma,F(Y)\otimes A),(\top,B)) \arrow[r, "\kappa_4"] \arrow[d] \arrow[dr, phantom, "(5)"] & \mathsf{Hom}_{Gr(E)}((\Gamma,F(Y)\otimes A),(\top,C)) \arrow[d] \\
\mathsf{Hom}_{E(\Gamma)}(F(Y)\otimes A,B) \arrow[r, "g_*"] & \mathsf{Hom}_{E(\Gamma)}(F(Y)\otimes A,C)
\end{tikzcd}$$

where κ_1 is the map which sends a morphism $(\langle\rangle, f)$ to the map

$$(\Gamma\times Y, A) \xrightarrow{(\pi,f)} (\Gamma, B) \xrightarrow{(\langle\rangle,g)} (\top, C)\ ,$$

κ_2 is the map which sends a morphim f to the morphism

$$F(\Gamma\times Y)\otimes A \xrightarrow{\hat{F}((\Delta\times\mathsf{Id}),\mathsf{Id})} F(\Gamma\times\Gamma\times Y)\otimes A \xrightarrow{m_{\hat{F}}^{-1}} F(\Gamma)\otimes F(\Gamma\times Y)\otimes A \xrightarrow{\mathsf{Id}\otimes f} F(\Gamma)\otimes B \xrightarrow{\hat{\nu}((\langle\rangle,g))} C\ ,$$

κ_3 is the map which sends a morphism f to the morphism

$$\begin{tikzcd}
F(\Gamma)\otimes F(Y)\otimes A \arrow[r, "\hat{F}((\Delta,\mathsf{Id}))\otimes\mathsf{Id}"] & F(\Gamma\times\Gamma)\otimes F(Y)\otimes A\otimes\mathsf{Id} \arrow[r, "m_{\hat{F}}^{-1}"] & F(\Gamma)\otimes F(\Gamma)\otimes F(Y)\otimes A \arrow[d, "\mathsf{Id}\otimes f"] \\
& C & F(\Gamma)\otimes B\ , \arrow[l, "\hat{\nu}((\langle\rangle,g))"']
\end{tikzcd}$$

and κ_4 is the map which sends a morphism $(\langle\rangle, f)$ to the morphism

$$(\Gamma, F(Y)\otimes A) \xrightarrow{(\mathsf{Id}, f)} (\Gamma, B) \xrightarrow{(\langle\rangle, g)} (\top, C)$$

It is easy to see that square (1) commutes. To show that square (2) commutes, consider the commuting diagram

$$\begin{tikzcd}
F(\Gamma \times Y)\otimes A \arrow[r, "\hat{F}((\Delta\times\mathsf{Id}),\mathsf{Id})"] \arrow[dd, "\mathsf{Id}"] & F(\Gamma \times \Gamma \times Y)\otimes A \arrow[r, "m_{\hat{F}}^{-1}"] \arrow[d, "\hat{F}((\mathsf{Id}\times(\langle\rangle,f)))"] & F(\Gamma)\otimes F(\Gamma \times Y)\otimes A \arrow[d, "\mathsf{Id}\otimes\hat{F}((\langle\rangle,f))"] \\
 & F(\Gamma \times \top)\otimes B \arrow[r, "m_{\hat{F}}^{-1}"] \arrow[dr, "\hat{F}((\pi,\mathsf{Id}))"] & F(\Gamma)\otimes F(\top)\otimes B \arrow[d, "\mathsf{Id}\otimes\hat{\epsilon}"] \\
F(\Gamma \times Y)\otimes A \arrow[rr, "\hat{F}((\pi,f))"] & & F(\Gamma)\otimes B \arrow[d, "\hat{\nu}((\langle\rangle,g))"] \\
 & & C\ ,
\end{tikzcd}$$

where the east-south-route is equal to $\kappa_2(\hat{\nu}((\langle\rangle, f)))$ and the south-east-route is equal to $\hat{\nu}(\kappa_1((\langle\rangle, f)))$.

The square (3) commutes because $m_{\hat{F}}^{-1}$ is a natural transformation.

To see that square (4) commutes, consider the commuting diagram in Figure 3, where for a morphism (f, g) in $Gr(E)$, the notation $(f, g)_2$ denotes the morphism g:

where the east-south-route is equal to $\hat{\nu}^{-1}(\kappa_3(f))$ and the south-east-route is equal to $\kappa_4(\hat{\nu}^{-1}(f))$.

To show that this adjunction satisfies the Beck-Chevally-condition, consider the following diagram, where in order to disambiguate (contrary to the rest of the paper) we write explicitly $E(f)$ instead of f^* if f is a morphism in $\mathcal{B}$ and use the notation g^* for the function on hom-sets which sends a

$$
\begin{tikzcd}
(\Gamma, F(Y)\otimes A) \arrow[r, "\hat{\eta}"] \arrow[d, "(\mathsf{Id}, r^{-1})"] & (\top, F(\Gamma)\otimes F(Y)\otimes A) \arrow[r, "(\mathsf{Id}, \hat{F}((\Delta, \mathsf{Id})))"] & (\top, F(\Gamma\times\Gamma)\otimes F(Y)\otimes A) \arrow[r, "(\top, m_{\hat{F}}^{-1})"] & (\top, F(\Gamma)\otimes F(\Gamma)\otimes F(Y)\otimes A) \arrow[d, "(\mathsf{Id}, r^{-1})"] \\
(\Gamma, F(Y)\otimes A\otimes I) \arrow[r, "\hat{\eta}"] \arrow[d, "(\mathsf{Id}, \hat{\nu}^{-1}(f)_2)\otimes\mathsf{Id}"] \arrow[dr, "(\Delta, \mathsf{Id})"] & (\top, F(\Gamma)\otimes F(Y)\otimes A\otimes I) \arrow[r, "(\mathsf{Id}, \hat{F}((\Delta, \mathsf{Id})))"] & (\top, F(\Gamma\times\Gamma)\otimes F(Y)\otimes A\otimes I) \arrow[r, "(\top, m_{\hat{F}}^{-1})"] & (\top, F(\Gamma)\otimes F(\Gamma)\otimes F(Y)\otimes A\otimes I) \arrow[d, "(\mathsf{Id}, r)"] \\
(\Gamma, B\otimes I) \arrow[ddr, "(\mathsf{Id}, r)"] & \begin{array}{c}(\Gamma\times\Gamma, F(Y)\otimes A\otimes I)\\ = (\Gamma, F(Y)\otimes A)\otimes(\Gamma, I)\end{array} \arrow[ur, "\hat{\eta}"] \arrow[r, "\hat{\eta}\otimes\hat{\eta}"] \arrow[d, "\hat{\nu}^{-1}(f)\otimes\mathsf{Id}"] & (\top, F(\Gamma)\otimes F(Y)\otimes A)\otimes(\top, F(\Gamma)\otimes I) \arrow[r, "(\langle\rangle, \sigma\cdot r)"] \arrow[d, "(\mathsf{Id}, f)\otimes\mathsf{Id}"] & (\top, F(\Gamma)\otimes F(\Gamma)\otimes F(Y)\otimes A) \arrow[d, "(\mathsf{Id}, \mathsf{Id}\otimes f)"] \\
& (\top, B)\otimes(\Gamma, I) \arrow[r, "\mathsf{Id}\otimes\hat{\eta}"] \arrow[d, "(\pi_2, r)"] & (\top, B)\otimes(\top, F(\Gamma)\otimes I) \arrow[r, "(\langle\rangle, r)"] & (\top, F(\Gamma)\otimes B) \arrow[d, "(\mathsf{Id}, \hat{F}((\langle\rangle, g)))"] \\
& (\Gamma, B) \arrow[urr, "\eta"] \arrow[d, "(\langle\rangle, g)"] & & (\top, F(\top)\otimes C) \arrow[d, "(\mathsf{Id}, \hat{\epsilon})"] \\
& (\top, C) \arrow[urr, "\hat{\eta}"] \arrow[rr, "\mathsf{Id}"] & & (\top, C)
\end{tikzcd}
$$

Figure 3: Commuting diagram for square (5)

morphism f to $g \circ f$.

$$
\begin{tikzcd}
\mathsf{Hom}_{E(\Gamma\times Y)}(A,B) \arrow[r, "E(g\times\mathsf{Id})"] \arrow[d] \arrow[dr, phantom, "(1)"] & \mathsf{Hom}_{E(\Delta\times Y)}(A,B) \arrow[d] \\
\mathsf{Hom}_{Gr(E)}((\Gamma\times Y,A),(\top,B)) \arrow[r, "(g\times\mathsf{Id},\mathsf{Id})^*"] \arrow[d] \arrow[dr, phantom, "(2)"] & \mathsf{Hom}_{Gr(E)}((\Delta\times Y,A),(\top,B)) \arrow[d] \\
\mathsf{Hom}_{E(\top)}(F(\Gamma\times Y)\otimes A,B) \arrow[r, "\hat{F}((g\times\mathsf{Id},\mathsf{Id}))^*"] \arrow[d] \arrow[dr, phantom, "(3)"] & \mathsf{Hom}_{E(\top)}(F(\Delta\times Y)\otimes A,B) \arrow[d] \\
\mathsf{Hom}_{E(\top)}(F(\Gamma)\otimes F(Y)\otimes A,B) \arrow[r, "\hat{F}((g,\mathsf{Id}))^*"] \arrow[d] \arrow[dr, phantom, "(4)"] & \mathsf{Hom}_{E(\top)}(F(\Delta)\otimes F(Y)\otimes A,B) \arrow[d] \\
\mathsf{Hom}_{Gr(E)}((\Gamma,F(Y)\otimes A),(\top,B)) \arrow[r, "(g,\mathsf{Id})^*"] \arrow[d] \arrow[dr, phantom, "(5)"] & \mathsf{Hom}_{Gr(E)}((\Delta,F(Y)\otimes A),(\top,B)) \arrow[d] \\
\mathsf{Hom}_{E(\Gamma)}(F(Y)\otimes A,B) \arrow[r, "E(g)"] & \mathsf{Hom}_{E(\Delta)}(F(Y)\otimes A,C)
\end{tikzcd}
$$

A direct calculation shows that square (1) commutes. Square (2) commutes because of the naturality of the adjunction defined in Lemma 29. The commutation of square (3) follows from the fact that the adjunction defined in Lemma 29 is monoidal. Square (4) commutes because of the naturality of the adjunction defined in Lemma 29. The commutation of square (5) can be shown by an easy calculation. □

9.2 The fibrational semantics for DILL

In this subsection we present the fibrational semantics for DILL. The difference between ILT and DILL is reflected also in the semantics: the primitive intuitionistic implication is replaced by the !-operator. Correspondingly the right adjunction to weakening which models the intuitionistic implication is replaced by an internalisation of the intuitionistic types. The details are as follows:

Definition 31 *A* DILL-indexed category *is a simply linear indexed category* $E\colon \mathcal{B}^{op} \to \mathbf{Cat}$ *such that the following conditions are satisfied:*

(i) Modalities. *For every object* A *of* $E(\top)$ *there exists an object* $!(A)$*, such that for all objects* Γ *of* $\mathcal{B}$ *the functor* $\mathsf{Fst}_A^*\colon E(\Gamma) \to E(\Gamma.A)$ *has a left adjoint* $!_A$ *such that* $!_A(B) = !(A) \otimes B$*. If* $f\colon B \to C$ *is a morphism in* $E(\Gamma.A)$*, we write* $\nu_A(f)$ *for the morphism in* $E(\Gamma)$ *obtained by transposing* f *across the adjunction.*

Moreover, the Beck-Chevalley condition for the adjunction $!_A \vdash \mathsf{Fst}_A^*$ *is satisfied in the strict sense, i.e. the equations*

$$f^* \cdot !_A = !_A \cdot (f.\mathsf{Id})^* \qquad f^*(\nu_A(g)) = \nu_A((f.\mathsf{Id})^*(g))$$

hold for every $f\colon \Delta \to \Gamma$*,* $B \in ObE(\Gamma)$*,* $C \in ObE(\Gamma)$ *and* $g \in \mathrm{Hom}_{E(\Gamma.A)}(B, C)$*.*

In addition we have for each morphism $f\colon B \to C$ *in* $E(\Gamma.A)$

$$\nu_A(f \otimes \mathsf{Id}) = \nu_A(f) \otimes \mathsf{Id}$$

(ii) Contextual base. *The objects of the category* $\mathcal{B}$ *are either* $\top_{\mathcal{B}}$ *or* $\top_{\mathcal{B}}.A_1.A_2. \ldots .A_n$ *where each* A_i *from* $i = 1, \ldots, n$ *is an object of* $E_{\top_{\mathcal{B}}}$*.*

Note that the restriction of the obects of the base category (condition (ii)) ensures that it suffices to require a left adjoint only to the functor Fst, as opposed to requiring a left adjoint to a more general projection functor in section 9.

Lemma 32 *In any DILL-indexed category* $E\colon \mathcal{B}^{op} \to \mathbf{Cat}$*,* $\mathcal{B}$ *is a cartesian category.*

Proof. The product of two objects $\top_{\mathcal{B}}.A_1 \cdots A_n$ and $\top_{\mathcal{B}}.B_1 \cdots B_m$ is the object $\top_{\mathcal{B}}.A_1 \cdots A_n.B_1 \cdots B_m$. The projections and product morphism can easily be defined by using the adjunction $U \dashv P$. $\square$

As for LNL, in DILL the intuitionistic implication $A \to B$ can be translated as $!A \multimap B$ whose semantic counterpart has the same proof as for proposition 21:

Proposition 33 *Consider a DILL-indexed category* $E\colon \mathcal{B}^{op} \to \mathbf{Cat}$*. For every object* Γ *of* $\mathcal{B}$ *and* A *of* $E(\Gamma)$*, the functor* $\mathsf{Fst}_A^*\colon E(\Gamma) \to E(\Gamma.A)$ *has a right adjoint* $\Pi_A\colon E(\Gamma.A) \to E(\Gamma)$*.*

The common part of DILL and ILT is reflected in the common structure of an ILT-category and a DILL-indexed category. Hence, the the interpretation of DILL in a DILL-indexed category and the interpretation of ILT in an ILT-category coincide. The parts of the syntax which are different are interpreted as follows (assuming that f^* on objects is the identity):

Definition 34 *Given any DILL-indexed category $E\colon \mathcal{B}^{op} \to \mathbf{Cat}$, we define a map $[\![\_]\!]$ from types to objects in $E(\top_{\mathcal{B}})$, from intuitionistic contexts Γ to objects in $\mathcal{B}$, from linear contexts Δ to objects in $E(\top_{\mathcal{B}})$, from double contexts to objects of suitable fibres and from terms $\Gamma \mid \Delta \vdash M\colon A$ to morphisms $[\![M]\!]\colon [\![\Delta]\!] \to [\![A]\!]$ in $E([\![\Gamma]\!])$ as follows:*

(i) For all constructors of ILT except $\to$, the interpretation of the corresponding constructors in DILL is given by the clauses for ILT;

(ii) On types: $[\![!A]\!] = !([\![A]\!])$

(iii) On terms:

$$\frac{[\![\Gamma \mid \_ \vdash M : A]\!] = h}{[\![\Gamma \mid \_ \vdash !M : !A]\!] = \langle \mathsf{Id}, h\rangle * (\eta_{[\![A]\!]}(I))}$$

$$\frac{[\![\Gamma | \Delta_1 \vdash M\colon !A]\!] = f \qquad [\![\Gamma, x\colon A | \Delta_2 \vdash N\colon B]\!] = g}{[\![\Gamma | \Delta \vdash_{\mathcal{L}} \mathsf{let}\ M\ \mathsf{be}\ !x\ \mathsf{in}\ N\colon B]\!] = \nu(g) \cdot (f \otimes \mathsf{Id}_{\Delta_2}) \cdot c}$$

where c is the canonical morphism from $[\![\Delta]\!]$ to $[\![\Delta_1]\!] \otimes [\![\Delta_2]\!]$ and η is the unit of the adjunction of definition 31.

The substitution lemmata for ILT carry over to DILL, and hence also the soundness theorem for ILT carries directly over to DILL:

Theorem 35 *Given a DILL-indexed category $E\colon \mathcal{B}^{op} \to \mathbf{Cat}$ under the above interpretation $[\![]\!]$ the following facts hold.*

(i) Assume $\Gamma \mid \Delta \vdash M\colon A$. Then $[\![\Gamma \mid \Delta \vdash M\colon A]\!]$ is a morphism from $[\![\Delta]\!]$ to $[\![A]\!]$ in $E([\![\Gamma]\!])$;

(ii) Assume $\Gamma \mid \Delta \vdash M = N\colon A$. Then $[\![\Gamma \mid \Delta \vdash M\colon A]\!] = [\![\Gamma \mid \Delta \vdash N\colon A]\!]$.

Proof. Routine induction over the structure of DILL-expressions using the substitution lemmas for DILL. □

DILL is also the internal language of DILL-indexed categories. The development for ILT carries through in exactly the same way. Firstly, we define the category of DILL-indexed categories:

Definition 36 *The category* **DILL-ind** *is the subcategory of* **SL-ind** *having as objects DILL-indexed categories and as morphisms between $E\colon \mathcal{B}^{op} \to$* **Cat** *and $E'\colon \mathcal{B}'^{op} \to$* **Cat** *are morphisms (H, α) in* **SL-ind** *such that for every object A of $E(top)$ and Γ of $\mathcal{B}$, the functors $\alpha_{\Gamma.A}$ and α_Γ are the components of a transformation of adjoints.*

Now, the fact that DILL is the internal language of DILL-indexed categories can be stated where that category TH(DILL) of DILL-theories is defined analogously to definition 15:

Proposition 37 *There exist two functors $L : \mathsf{TH}(DILL) \to$ **DILL-ind** and $F :$ **DILL-ind** $\to \mathsf{TH}(DILL)$ that give rise to an equivalence between $\mathsf{TH}(DILL)$ and* **DILL-ind**.

Analogously, we can define **DILL$_p$-ind** as the subcategory of **DILL-ind** having as objects those indexed categories whose fibres are SMCC with finite products and whose projecting functors preserve them and having as morphisms those of **DILL-ind** whose natural transformations are componentwise SMCP functors. Then, we can prove that $_{\&}$DILL provides the internal languages for the indexed categories in **DILL$_p$-ind**.

There is also an alternative definition of a DILL-category which in the same way as in the LNL-case requires a left adjoint to the inclusion functor between the base category modelling intuitionistic terms and the category $E(\top)$ which models all the linear terms.

Proposition 38 *Let $E\colon \mathcal{B}^{op} \to$ **Cat** be a simple indexed category. Then the following are equivalent:*

(i) E is an DILL-category;

(ii) The functor $\mathcal{I}_{\mathcal{B}}\colon E(\top) \to \mathcal{B}$ which maps an object A to $\top.A$ and a morphism f to $\mathsf{Id}.f$ has a left adjoint $F_{\mathcal{B}}$, and this adjunction is monoidal.

Proof. The proof of section 9 carries over directly. □

10 Equivalence between fibrational models and models based on monoidal adjunctions

In this section we show that the models for LNL and DILL defined in the previous sections are equivalent to the ones given in the literature.

The models in the literature model the intuitionistic part of the type theory by a cartesian closed category and the linear part by one symmetric monoidal closed category, contrary to the families of symmetric monoidal closed categories in the fibration models discussed in this paper. The interaction between intuitionistic and linear type theory is modelled by an adjunction between the cartesian closed category and the symmetric monoidal closed category. Part of this adjunction are functors between the cartesian closed category and symmetric monoidal closed category. These functors give rise to an endofunctor on the symmetric monoidal closed categories which models the exponentials. Hence this framework is too strong to provide a model for ILT, and so we consider in this section only models for DILL and LNL.

We start by relating our categorical models for LNL to the ones in the literature. The definition of a model for LNL is as follows:

Definition 39 (Benton) *A* LNL-model *consists of a symmetric monoidal adjunction $K \dashv J$ with $K: \mathbb{C} \to \mathbb{S}$, $J: \mathbb{S} \to \mathbb{C}$ where $\mathbb{S}$ is a symmetric monoidal category and $\mathbb{C}$ is a cartesian closed category.*

The equivalence theorem relates LNL-models and LNL-indexed categories. Its precise statement is an equivalence between the corresponding categories of models. In [MMdPR01], we investigate the relationships between the models for various linear type theories more closely; here we cite only the relevant definitions and results.

The categories of models for the various linear type theories considered in this paper are all categories of monoidal adjunctions between a cartesian category and a symmetric monoidal category. We denote these categories by **MA** (for monoidal adjunction) with added superscripts and subscripts to indicate additional structure.

For LNL, this additional structure is the intuitionistic function spaces, and hence we denote the category of LNL-models by $^{\to}\mathbf{MA}$. This category is defined as follows:

Definition 40 *The category $^{\to}\boldsymbol{MA}$ (resp. $^{\to}\boldsymbol{MA}_{st}$) has as objects LNL-models. A morphism between two LNL-models, i.e.,symmetric monoidal adjunctions $K \dashv J, K' \dashv J'$, is a pair of functors (P, H) where $K: \mathbb{S} \to \mathbb{S}'$ is a strong (resp. strict) symmetric monoidal closed functor and $H: \mathbb{C} \to \mathbb{C}'$ is a cartesian closed (resp. strict cartesian closed) functor such that (P, H) forms a* monoidal *transformation of adjoints.*

We call $^{\to}_{\&}\mathbf{MA}$ ($^{\to}_{\&}\mathbf{MA}_{st}$) the category of monoidal adjunctions in $^{\to}\mathbf{MA}$ ($^{\to}\mathbf{MA}_{st}$) having as objects those in $^{\to}\mathbf{MA}$ whose SMCC has finite products and as morphisms those in $^{\to}\mathbf{MA}$ ($^{\to}\mathbf{MA}_{st}$) whose SMC component preserves (strictly) also the finite products.

Now we turn to the equivalence between fibrational models for LNL and $\mathbf{MA}_{st}$. The structure of an LNL-model is already contained in a fibrational model for LNL: in any LNL-indexed category, there is already a monoidal adjunction between a cartesian closed category (namely the base category $\mathcal{B}$) and a symmetric monoidal closed category $E(\top_B)$. This provides one direction of the equivalence. The other direction of the equivalence is more complicated, as one has to construct an indexed category which models the indexing over intuitionistic contexts out of a monoidal adjunction between a cartesian closed category $\mathbb{C}$ and a symmetric monoidal closed category $\mathbb{S}$. It turns out that the correct choice is to define the base category to be $\mathbb{C}$ and to define the fibres over an object Γ of $\mathbb{C}$ to be the co-Kleisli category with respect to the functor $K(\Gamma) \otimes \_$. By the adjunction between $\mathbb{C}$ and $\mathbb{S}$ the morphisms of such a fibre model terms with free intuitionistic variables are given by the context Γ.

The precise statement of the equivalence is as follows:

Proposition 41 *The category $\boldsymbol{MA}_{st}$ is equivalent to the category* **LNL-ind**.

Proof. Observe that any symmetric monoidal adjunction $K \vdash J$, with $K : \mathcal{C} \to \mathcal{S}$, $\mathcal{C}$ a cartesian closed category and $\mathcal{S}$ a symmetric monoidal closed category provides an IL-indexed category, by taking, as a base, the cartesian closed category $\mathcal{C}$ and, as the fibre over an object C of the base the symmetric monoidal closed category whose objects are those of $\mathcal{S}$ and whose morphisms from A to B are the $\mathcal{S}$-morphisms from $K(C) \otimes A$ to B. The modality condition is obviously satisfied. Finally, the comprehension functor is defined by putting $G((\Gamma, A)) = \Gamma \times K(A)$ and $G((f, h)) =< f \cdot \pi_1, K(h) \cdot m_K \cdot (\eta \times id) >$ where η is the unit of the adjunction, m_K is the monoidal natural transformation of K and π_1 the first projection of the binary product.

Conversely, any indexed category $E : \mathcal{B} \to \mathbf{Cat}$ in **LNL-ind** gives rise to a monoidal adjunction: the symmetric monoidal closed category is $E(\top_{\mathcal{B}})$, and the cartesian closed category is the base category $\mathcal{B}$.

Using Proposition 27 one shows that two LNL-indexed categories which have isomorphic base categories and fibres over the terminal object are isomorphic. Now it is easy to show that the translations from an LNL-model to an LNL-indexed category and vice versa, which can be lifted up to the corresponding functors, yield the desired equivalence. $\square$

In previous sections we showed that the indexed models for DILL were restrictions of the indexed models for LNL where the cartesian closed base category is generated as the free cartesian closed category on the co-Kleisli category associated to the adjunction. Hence, the cartesian closed category is determined by the symmetric monoidal closed fibres. Similarly, DILL-models arise as restrictions of LNL-models such that the cartesian closed category is a free structure determined by the symmetric monoidal closed category. In [MMdPR01] we investigate which additional requirements on a symmetric monoidal closed category are necessary to ensure the existence of this free structure. It turns out that a suitable definition is as follows:

Definition 42 *A* DILL-model *is a symmetric monoidal category $\mathbb{S}$ together with an endofunctor $!$ such that the monoidal structure $(\otimes^!, I^!)$ of the category of (coEilenberg-Moore) coalgebras $\mathbb{S}^!$ is a finite product structure.*

In [MMdPR01] we also show that each DILL-model gives uniquely rise to a monoidal adjunction $F^! \dashv K^!$ with $F^! : \mathbb{S}^p_k \to \mathbb{S}$, where $\mathbb{S}^p_k$ is the category of finite products of free coalgebras in $\mathbb{S}$ and $F^! \dashv K^!$ is the restriction of the canonical adjunction of the co-Eilenberg-Moore construction. In particular $F^! : \mathbb{S}^p_k \to \mathbb{S}$ is the functor sending the finite product of coalgebras to their support and $K^! : \mathbb{S} \to \mathbb{S}^p_k$ is the embedding functor of $\mathbb{S}$ in the category $\mathbb{S}^p_k$.

We now use these monoidal adjunctions to define the category of DILL-models as a subcategory of LNL-models:

Definition 43 *$\boldsymbol{MA}^{DILL}$ ($\boldsymbol{MA}^{DILL}_{st}$) is the full subcategory of $\boldsymbol{MA}$ ($\boldsymbol{MA}_{st}$) having as objects the monoidal adjunctions corresponding to DILL-models.*

We call ${}_{\&}\mathbf{MA}^{\mathrm{k}}$ (${}_{\&}\mathbf{MA}^{\mathrm{k}}_{st}$) the subcategory of monoidal adjunctions in $\mathbf{MA}^{\mathrm{DILL}}$ ($\mathbf{MA}^{\mathrm{DILL}}_{st}$) having as objects those in $\mathbf{MA}^{\mathrm{DILL}}$ whose SMCC has finite products and as morphisms those in $\mathbf{MA}^{\mathrm{DILL}}$ ($\mathbf{MA}^{\mathrm{DILL}}_{st}$) whose SMC component preserves (strictly) also the finite products.

Analogously, we prove that category **DILL-ind** of fibrational models for DILL is equivalent to the category $\mathbf{MA}^{\mathrm{DILL}}$ of DILL-models in term of monoidal adjunctions defined in [MMdPR01] as a restriction of Benton's models in [Ben95]. This proof can be obtained by observing that the restrictions which lead from an LNL-model to a DILL-model and from an LNL-indexed category to a DILL-indexed category match.

Proposition 44 *The category $\boldsymbol{MA}^{DILL}_{st}$ is equivalent to the category* **DILL-ind**.

Proof. The proof is exactly the same as for proposition 41. Just note that given a monoidal adjunction in $\mathbf{MA}^{\mathrm{DILL}}_{st}$ the finite products co-Kleisli category related to a monoidal comonad is exactly the cartesian category generated by the cofree coalgebras and each cofree coalgebra corresponds to one object of the fibre on empty context turned into the base category via comprehension. $\square$

Now, having proved quite naturally that DILL and LNL respectively provide the internal language of the corresponding fibrational models, from the above equivalences in proposition 41 and proposition 44 we get another proof that DILL and LNL respectively provide the internal language for $\mathbf{MA}^{\mathrm{DILL}}_{st}$ and ${}^{\rightarrow}\mathbf{MA}_{st}$.

Clearly, all this still holds if we add products, namely that **DILL$_p$-ind** is equivalent to ${}_{\&}\mathbf{MA}^{\mathrm{k}}_{st}$ and that **LNL$_p$-ind** is equivalent to ${}^{\rightarrow}_{\&}\mathbf{MA}_{st}$.

11 How to add exponentials to ILT

The Girard-translation shows that ILT can be seen as a subcalculus of DILL. In this section we develop this view further and show that DILL is a conservative extension of ILT.

We start by showing that the free addition of the modality ! to ILT leads to DILL. Categorically, this is expressed by the existence of a left adjoint to the inclusion functor of DILL-indexed categories into the category of ILT-indexed categories

$$\mathcal{I}: \ \mathbf{DILL\text{-}ind} \hookrightarrow \mathbf{ILT\text{-}ind}$$

so that **DILL-ind** becomes a reflective subcategory of **ILT-ind**:

Proposition 45 *The inclusion functor* $\mathcal{I}$: $\mathbf{DILL\text{-}ind} \hookrightarrow \mathbf{ILT\text{-}ind}$ *has a left adjoint*

$$S : \mathbf{ILT\text{-}ind} \longrightarrow \mathbf{DILL\text{-}ind}$$

Proof. A way to describe the functor S is to use the internal language of ILT-indexed categories: given any ILT-indexed category E, we freely add the ! modality to the internal language of E, i.e. we get a DILL-theory, and then we define $S(E)$ to be the syntactic ILT-indexed category associated with this extended language, which happens to have exponentials. This addition of exponentials is a free construction, hence the functor S is a left adjoint. □

Note that the functor $\mathcal{I}$ is in effect the Girard-translation: the intuitionistic function space $A \to B$ in the ILT-indexed category $\mathcal{I}(E)$ is given by $!A \multimap B$ in the DILL-indexed category E.

This result only shows that the exponentials can be added to ILT without affecting the already existing structure. However, ILT admits already a limited kind of exponentials. This can be seen as follows, using the Grothendieck completion. Indeed, if we look closely at the categorical structure of the Grothendieck completion and its comprehension adjunction in the case of an ILT-indexed category, we see that it gives rise to a Benton model.

Proposition 46 *For any ILT-indexed category* $E\colon \mathcal{B}^{op} \to \mathbf{Cat}$*, the adjunction* $U \dashv P$ *with* $U : \mathcal{B} \to Gr(E)$ *is a Benton model, i.e.* $Gr(E)$ *is a symmetric monoidal closed category,* $\mathcal{B}$ *a cartesian closed one and* $U \dashv P$ *a symmetric monoidal adjunction.*

Proof. We recall that $\mathcal{B}$ is a cartesian closed category, as already noticed by means of the internal language. We then describe the symmetric monoidal structure of $Gr(E)$: the tensor and the linear function space are respectively, assuming for simplicity that, for every base morphism f, the functor f^* is the identity on objects

$$\begin{array}{l}(\Gamma, A) \otimes (\Delta, B) = (\Gamma \times \Delta, A \otimes B) \\ (\Gamma, A) \multimap (\Delta, B) = (\Gamma \to \Delta, \Pi_{\tilde{\Gamma}}(A \multimap B))\end{array}$$

where $\tilde{\Gamma}$ is the object such that $\top_{\mathcal{B}}.\tilde{\Gamma} = \Gamma$. If $\Gamma = \top_{\mathcal{B}}$ then $(\top_{\mathcal{B}}, A) \multimap (\Delta, B) \cong (\top_{\mathcal{B}} \to \Delta, A \multimap B)$.

In order to prove the closure of $Gr(E)$, i.e. for example that

$$\mathsf{Hom}_{Gr(E)}((\Gamma \times \Gamma', A \otimes A') \,,\, (\Delta, B)) \cong \mathsf{Hom}_{Gr(E)}((\Gamma, A) \,,\, (\Gamma' \to \Delta, \Pi_{\tilde{\Gamma'}}(A' \multimap B)))$$

we use the closure of the fibres and the right adjoint to the weakening functor as follows

$$\begin{array}{rcl}\mathsf{Hom}_{E(\Gamma \times \Gamma')}(A \otimes A', B) & \cong & \mathsf{Hom}_{E(\Gamma \times \Gamma')}(A, A' \multimap B) \\ & \cong & \mathsf{Hom}_{E(\Gamma.\tilde{\Gamma'})}(A, A' \multimap B) \\ & \cong & \mathsf{Hom}_{E(\Gamma)}(A, \Pi_{\tilde{\Gamma'}}(A' \multimap B))\end{array}$$

where we recall that by the internal language we can prove $\Gamma \times \Gamma' \cong \Gamma.\tilde{\Gamma'}$. Moreover, the functor U is strongly monoidal since

$$U(\Gamma \times \Delta) \cong (\Gamma \times \Delta, I) \simeq (\Gamma, I) \otimes (\Delta, I)$$

Hence the adjunction happens to be symmetric monoidal.

In conclusion the above adjunction gives rise to a Benton model, and even more to a model in $\mathbf{MA}^{\mathrm{DILL}}$, since the category $\mathcal{B}$ is isomorphic to the co-Kleisli category of the considered adjunction. $\square$

Note that the Benton-model described in the previous proposition is a particular one: the functor ! turns out to be defined by $!(\Gamma, A) = (\Gamma \times A, I)$, and hence $!!(\Gamma, A) = !(\Gamma, A)$. This means that this Benton-model has an idempotent co-monad, and as consequence the cartesian category is simply the full subcategory of all objects $!A$ for some object A. This fact is reflected in this model by the cartesian category $\mathcal{B}$ being the first component of the Grothendieck-construction. Moreover, as any object of the base category $\mathcal{B}$ is of the form $\top \cdot A_1 \cdots A_n$, where the objects A_i are objects of $E(\top)$, this Benton-model gives rise also to a categorical model of DILL. The type-theoretic intuition behind this model is as follows: Types $!A$ admit arbitrary weakening and contraction. In ILT a context $x\colon A$ fulfills the same role, and the above model formalises this intuition.

But it is not true that this model gives rise to an equivalence between DILL with the additional assumption that $!!A$ is isomorphic to $!A$ and ILT. The reason is the definition of the monoidal function spaces in the Grothendieck-construction: the linear function space of (A, B) and (C, D) is $(A \to C, A \to B \multimap D)$, but there is no dependency of $A \to C$ on B. Such a situation is possible in DILL as the type $A \multimap !B$ demonstrates.

The existence of types of the form $A \multimap !B$ is in fact the key difference between ILT and DILL. On the types which arise from the Girard-translation DILL and ILT coincide; in other words, DILL is a conservative extension of ILT. More precisely, we identify a set of DILL-types such that the range of the Girard-translation from ILT into DILL covers all the terms with these types, and that two terms M and N of these types are equal iff any two ILT-terms M' and N' such that $(M')^{DILL} = M$ and $(N')^{DILL} = N$ are equal. This means that DILL contains just extra types and their corresponding terms compared to ILT and coincides on the common types with ILT.

We show this conservativity result for a calculus ILT where we replace the type I by the terminal type 1. In DILL, I is isomorphic to $!1$, and this isomorphism suggests that I is in a fact a !-type and therefore should not be part of ILT. We call this calculus ILT1. To prove this conservativity result, we define a translation from a part of DILL to ILT1 and show that this translation is inverse to the Girard-translation.

This subcalculus of DILL is DILL where the types are only the ones given by the grammar

$$\begin{array}{rcl} L & ::= & G \mid L \otimes L \mid 1 \mid L \multimap L \mid In \multimap L \\ In & ::= & !L \end{array}$$

and all the contexts and terms that can be formed using these types only.

The translation from this subcalculus of DILL to ILT1 is defined as follows:

(i) On types:

$$\begin{array}{rcl} (G)^{ILT} & = & G \\ (L \otimes L')^{ILT} & = & (L)^{ILT} \otimes (L')^{ILT} \\ (1)^{ILT} & = & 1 \\ (L \multimap L')^{ILT} & = & (L)^{ILT} \multimap (L')^{ILT} \\ (In \multimap L)^{ILT} & = & (In)^{ILT} \to (L)^{ILT} \\ (!L)^{ILT} & = & (L)^{ILT} \end{array}$$

(ii) On contexts: Let $\Gamma = x_1 \colon A_1, \ldots, x_n \colon A_n$. Define

$$(\Gamma)^{ILT} = x_1 \colon (A_1)^{ILT}, \ldots, x_n \colon (A_n)^{ILT} \; .$$

Let $\Delta = a_1 \colon B_1, \ldots, a_m \colon B_m$, let Δ_1 be the context $a_1 \colon B_1, \ldots, a_k \colon B_k$ where the types in $B_1, \ldots, B_k$ are exactly the exponential types $!C_i$ for some type C_i and let Δ_2 be the context $a_{k+1} \colon B_{k+1}, \ldots, a_m \colon B_m$. Then $(\Delta_1)^{ILT} = x_{n+1} \colon (B_1)^{ILT}, \ldots, x_{n+k} \colon (B_k)^{ILT}$, and $(\Delta_2)^{ILT} = a_{k+1} \colon (B_{k+1})^{ILT}, \ldots a_n \colon , (B_n)^{ILT}$. Then

$$(\Gamma | \Delta_1, \Delta_2)^{ILT} = (\Gamma)^{ILT}, (\Delta_1)^{ILT} \mid (\Delta_2)^{ILT} \; .$$

(iii) On terms:

$$\begin{array}{rcl} (x)^{ILT} & = & x \\ (a)^{ILT} & = & \left\{ \begin{array}{ll} x & \text{if } a \text{ is a variable of type } !A \\ a & \text{otherwise} \end{array} \right. \\ (!M)^{ILT} & = & (M)^{ILT} \\ (\mathsf{let}\ M\ \mathsf{be}\ !x\ \mathsf{in}\ N)^{ILT} & = & (N)^{ILT}[(M)^{ILT}/x] \\ (\lambda a^A.M)^{ILT} & = & \left\{ \begin{array}{ll} \lambda x^{(B)^{ILT}}.(M)^{ILT} & \text{if } A =!B \\ \lambda a^{(A)^{ILT}}.(M)^{ILT} & \text{otherwise} \end{array} \right. \\ (MN)^{ILT} & = & \left\{ \begin{array}{ll} (M)^{ILT}_I (N)^{ILT} & \text{if } M \text{ is of type } !A \\ (M)^{ILT}_L (N)^{ILT} & \text{otherwise} \end{array} \right. \\ (M \otimes N)^{ILT} & = & (M)^{ILT} \otimes (N)^{ILT} \\ (\mathsf{let}\ M\ \mathsf{be}\ a \otimes b\ \mathsf{in}\ N)^{ILT} & = & \mathsf{let}\ (M)^{ILT}\ \mathsf{be}\ a \otimes b\ \mathsf{in}\ (N)^{ILT} \\ (\langle\rangle)^{ILT} & = & \langle\rangle \end{array}$$

Next we show that this translation preserves typing.

Lemma 47 *(i) If $\Gamma|\Delta \vdash M\colon !L$ in the subset of DILL, then Δ contains only exponential types.*

(ii) Assume $\Gamma|\Delta \vdash M\colon A$ in the subset of DILL. Then also $(\Gamma|\Delta)^{ILT} \vdash (M)^{ILT}\colon (A)^{ILT}$.

Proof. (i) We show this statement by induction over M. The important point is that an application MN cannot have a type $!L$, and hence this case is vacuously true.

(ii) Again, we show this statement by induction over M. We present here only the case of $\lambda a^{!A}.M$ and MN when M is of type $!A \multimap B$. In the first case, by induction hypothesis we have $(\Gamma|\Delta)^{ILT}, x^{(A)^{ILT}} \vdash (M)^{ILT}\colon (B)^{ILT}$, and hence also $(\Gamma|\Delta, a\colon !A)^{ILT} \vdash (M)^{ILT}\colon (B)^{ILT}$. So we obtain $(\Gamma|\Delta)^{ILT} \vdash (\lambda a^{!A}.M)^{ILT}\colon (!A \multimap B)^{ILT}$. In the second case, by induction hypothesis $(\Gamma|\Delta)^{ILT} \vdash (M)^{ILT}\colon (A)^{ILT} \to (B)^{ILT}$, and $(\Gamma|\Delta)^{ILT} \vdash (N)^{ILT}\colon (A)^{ILT}$, and N has no free linear variables. Hence we also have $(\Gamma|\Delta)^{ILT} \vdash (M)^{ILT}(N)^{ILT}\colon (B)^{ILT}$.

□

Before we can show preservation of equalities, we need a substitution lemma.

Lemma 48 *Consider terms M and N such that $\Gamma|\Delta_1 \vdash M\colon A$ and $\Gamma|\Delta_2, a\colon A \vdash N\colon B$. If A is not an exponential type, then $(M[N/a])^{ILT} = (M)^{ILT}[(N)^{ILT}/a]$. If A is an exponential type $!B$, then $(M[N/a])^{ILT} = (M)^{ILT}[(N)^{ILT}/x]$, where $(a)^{ILT} = x$.*

Proof. Simultaneous induction over M. The only interesting case is the one of a variable a of type $!L$. In this case we have

$$(a[N/a])^{ILT} = (N)^{ILT} = x[(N)^{ILT}/x] = (a)^{ILT}[(N)^{ILT}/x]$$

□

Now we can show the preservation of equalities.

Lemma 49 *Assume $\Gamma|\Delta \vdash M = N\colon A$. Then $(\Gamma|\Delta)^{ILT} \vdash (M)^{ILT} = (N)^{ILT}\colon (A)^{ILT}$.*

Proof. We have to consider each equality separately. The cases involving the tensor product and terminal object are immediate. Now consider the β-equation for !. We have $(\mathsf{let}\ !M\ \mathsf{be}\ !x\ \mathsf{in}\ N)^{ILT} = (N)^{ILT}[(M)^{ILT}/x]$, which is by the substitution lemma (Lemma 48) equal to $(N[M/x])^{ILT}$. Next, we consider the η-equality for !. We have

$$\begin{aligned}(\mathsf{let}\ M\ \mathsf{be}\ !x\ \mathsf{in}\ N[!x/a])^{ILT} &= (N)^{ILT}[(M)^{ILT}/x] \text{where } (a)^{ILT} = x \\ &= (N[M/a])^{ILT}\end{aligned}$$

where we use the substitution lemma (Lemma 48) in both steps.

Now we consider the β-equality. The only interesting case is the one of a term $(\lambda a^{!L}.M)N$. Here we have

$$\begin{array}{rcl}((\lambda a^{!L}.M)N)^{ILT} & = & (\lambda x^{(L)^{ILT}}.(M)^{ILT})(N)^{ILT} \text{ where } (a)^{ILT} = x \\ & = & (M)^{ILT}[(N)^{ILT}/x] \\ & = & (M[N/a])^{ILT}\end{array}$$

□

Now we show the conservativity result:

Theorem 50 *Assume $\Gamma|\Delta, \Delta' \vdash M \colon A$ in the subcalculus of DILL where Δ and Δ' contain only linear types and exponential types respectively and A is a linear type.*

(i) There exists a well-formed term M' in ILT1 such that $(M')^{DILL} = M$ if A is a linear type and $!(M')^{DILL} = M$ if $A = !L$ modulo the isomorphism of Proposition 2 between $\mathsf{Hom}((\Gamma|\Delta, a\colon !A), B)$ and $\mathsf{Hom}((\Gamma, x\colon A|\Delta), B)$;

(ii) if $M = N$ in DILL, then also $M' = N'$, where M' and N' are the terms from part (i).

Proof. The term M' is the term $(M)^{ILT}$. By Lemma 47, this term is a well-formed ILT-term. Now we show the statement by induction over the structure of the term M. We present here only some cases, the other ones are immediate. First, we consider the case of $\lambda a^{!L}.M$. In this case, we have

$$\begin{array}{rcl}((\lambda a^{!L}.M)^{ILT})^{DILL} & = & (\lambda x^{(L)^{ILT}}.(M)^{ILT})^{DILL} \text{ where } (a)^{ILT} = x \\ & = & \lambda a^{!L}.\mathsf{let}\ a\ \mathsf{be}\ !x\ \mathsf{in}\ ((M)^{ILT})^{DILL}\end{array}$$

Now the induction hypothesis yields

$$((\lambda a^{!L}.M)^{ILT})^{DILL} = \lambda a^{!L}.\mathsf{let}\ a\ \mathsf{be}\ !x\ \mathsf{in}\ M[!x/a]$$

modulo Φ (note that a is a variable of exponential type), and now the η-rule for ! yields

$$((\lambda a^{!L}.M)^{ILT})^{DILL} = \lambda a^{!L}.M$$

which was to be shown.

Next we present the case of an application MN, where M is of type $!L \multimap B$. In this case we have

$$\begin{array}{rcl}((MN)^{ILT})^{DILL} & = & ((M)^{ILT})^{DILL}(!((N)^{ILT})^{DILL}) \\ & = & MN\end{array}$$

Finally we consider the case $\mathsf{let}\ M\ \mathsf{be}\ !x\ \mathsf{in}\ N$. By Lemma 47, $(M)^{ILT}$ has no free linear variables, and $((M)^{ILT})^{DILL} = !M' = M$ modulo Φ. Hence modulo Φ we have $\mathsf{let}\ M\ \mathsf{be}\ !x\ \mathsf{in}\ N = \mathsf{let}\ !R\ \mathsf{be}\ !x\ \mathsf{in}\ N = N[R/x]$. Now the induction hypothesis and the substitution lemma 48 imply that $((N[R'/x])^{ILT})^{DILL} = N[R'/x]$. □

This result explains why for applications in functional programming ILT often suffices: if linearity is used to capture whether arguments may be duplicated or discarded, then there is no need for types of the form $A \multimap !B$, and hence ILT with its easier syntax suffices.

12 A summary of the relationships between the models

In previous sections we already mentioned relationships between ILT, DILL and LNL. In [MMdPR01] we studied the relationships between the non-fibrational categorical models for DILL and LNL in detail. As we have shown the equivalence between the categorical models using fibrations and without them, we obtain from this work the following picture describing the relationships between the various models, where the models in the bottom line are defined using fibrations and the models in the top line do not use fibrations:

$$\begin{tikzcd}
 & {}_{\&}\mathbf{Lin}_{st} \arrow[r, leftrightarrow, "\simeq"] & {}_{\&}\mathbf{MA}^{\mathrm{k}}_{st} \arrow[r, shift left, "I"] \arrow[r, phantom, "\perp"] \arrow[d, leftrightarrow, "\simeq"] & {}^{\rightarrow}_{\&}\mathbf{MA}_{st} \arrow[l, shift left, "R"] \arrow[d, leftrightarrow, "\simeq"] \\
\mathbf{ILT}_p\textbf{-ind} & & \mathbf{DILL}_p\textbf{-ind} \arrow[ll, "\mathcal{I}"] \arrow[r, shift left, "\mathcal{I}"] \arrow[r, phantom, "\top"] & \mathbf{LNL}_p\textbf{-ind} \arrow[l, shift left, "R"]
\end{tikzcd}$$

where ${}_{\&}\mathbf{Lin}_{st}$ is the subcategory of $\mathbf{Lin}_{st}$ having as objects those of $\mathbf{Lin}_{st}$ whose SMCC have also finite products and having as morphisms those of $\mathbf{Lin}_{st}$ that preserves strictly finite products. For the equivalence between $\mathbf{Lin}_{st}$ and $\mathbf{MA}^{\mathrm{DILL}}_{st}$ and between ${}_{\&}\mathbf{Lin}_{st}$ and ${}_{\&}\mathbf{MA}^{\mathrm{k}}_{st}$ see [MMdPR01].

This leaves only the question of a non-fibrational categorical model for ILT, which would be placed in the left upper corner of the previous picture. Proposition 46 indicates how such a model has to look like: in such a model there exists a functor ! such that $!!A$ is isomorphic to $!A$. However, we also show in the last section that this condition is not enough to characterise ILT, and that the image of the translation of ILT into DILL does not capture all DILL-types. Hence a definition of a non-fibrational categorical model for ILTseems to be problematic, and we regard the fibrational categorical models as the natural categorical models of ILT.

13 Conclusion

We have produced a sound and complete categorical semantics for the type theory ILT, a system that allows both intuitionistic and linear implication to coexist as primitive operations, without the modality !. The canonical categorical models are fibrational: ILT-indexed categories, in which the base category models intuitionistic contexts and the fibres, which are symmetric monoidal closed categories, model linear terms parametrised by those contexts. We have shown that, with a mild restriction on the indexed categories, ILT is the *internal language* of its models—a stronger result than soundness and completeness alone, expressing that the type theory and its categorical models are interchangeable.

Beyond ILT itself, we have provided fibrational semantics for DILL and LNL and established their equivalence with the classical monoidal-adjunction semantics of Barber–Plotkin and Benton respectively. The resulting picture is summarised in the diagram in Section 6: fibrational models sit at the bottom, their monoidal-adjunction counterparts at the top, and the horizontal relations record equivalences of categories of models.

A further contribution is the conservativity result: DILL is a conservative extension of ILT. The types that lie outside ILT are precisely those of the form $A \multimap !B$ and $!!A$; on the common fragment the two systems coincide. This explains why ILT suffices for the main applications of linear logic in functional programming, where the key information is whether an argument may be discarded or duplicated, and not whether it may be promoted to an arbitrary exponential.

Why fibrations are the right models for ILT. The conservativity result also explains why a clean non-fibrational categorical model for ILT is hard to formulate: the image of ILT in DILL does not generate all DILL-types, so there is no obvious way to carve out the ILT-fragment of a monoidal adjunction model. The fibrational models, by contrast, are the *natural* models: they respect the double-context structure $\Gamma \mid \Delta$ directly, rather than encoding it via the modality.

Relation to dependent type theory. The fibration framework used here is closely related to the comprehension categories and display-map categories used in the categorical semantics of dependent type theory [Jac99, Ehr88]. Indeed, the IL-indexed categories of this paper are fibred structures over a cartesian base, satisfying the comprehension axiom of Lawvere [Law70], and the fibres carry a linear (symmetric monoidal closed) rather than cartesian structure. This observation opens a natural path toward a dependent version of ILT: one would allow types in the fibres to depend on terms in the base, while keeping the linear discipline on fibre morphisms. Such a system would combine intuitionistic dependent types (modelled by the base) with linear types (modelled by the fibres). The semantic framework was already in place to support it since the conference paper [MdPR00].

Work on linear dependent type theory (see, e.g., Vákár [Vák17], and

Nuyts–Devriese [ND18]) and on graded or modal dependent type theories (Gratzer et al. [GKNB20]) proceeds from very similar motivations, often rediscovering the importance of a double-context discipline. We hope the semantic analysis presented here (mostly in work done alreay in 2000—in particular the central role of the comprehension adjunction and the fibred SMC structure—will be useful in that programme.

The correspondence between DILL-indexed categories and monoidal adjunctions also has a direct counterpart in the more recent literature on *linear HoTT* and *cohesive type theory* (Shulman [Shu18], Licata–Shulman [LS16]), where an adjunction between a cartesian and a linear (or cohesive) mode plays the same structural role as the adjunction $F \dashv G$ in LNL. Making this connection precise is a goal for future work.

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